

CamCAN Selectivity & Attrition Analysis (for *The Cambridge Centre for Ageing and Neuroscience (Cam-CAN) longitudinal study protocol: Phase 4 (Enrichment) and Phase 5 (Rescan)*)

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```
library(ggplot2)
library(ggpubr)
library(gridExtra)
library(grid)
library(cowplot)
```

```
##
## Attaching package: 'cowplot'

## The following object is masked from 'package:ggpubr':
##
##   get_legend
```

```
library(GGally)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following object is masked from 'package:gridExtra':
##
##   combine

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(patchwork)
```

```
##
## Attaching package: 'patchwork'
```

```
## The following object is masked from 'package:cowplot':  
##  
##   align_plots
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v forcats   1.0.1      v stringr   1.6.0  
## v lubridate 1.9.5      v tibble   3.3.1  
## v purrr     1.2.2      v tidyr    1.3.2  
## v readr     2.2.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::combine()   masks gridExtra::combine()  
## x dplyr::filter()    masks stats::filter()  
## x dplyr::lag()       masks stats::lag()  
## x lubridate::stamp() masks cowplot::stamp()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(isotree)  
library(lmerTest)
```

```
## Loading required package: lme4  
## Loading required package: Matrix  
##  
## Attaching package: 'Matrix'  
##  
## The following objects are masked from 'package:tidyr':  
##  
##   expand, pack, unpack  
##  
## Registered S3 method overwritten by 'lme4':  
##   method           from  
##   na.action.merMod car  
##  
## Attaching package: 'lmerTest'  
##  
## The following object is masked from 'package:lme4':  
##  
##   lmer  
##  
## The following object is masked from 'package:stats':  
##  
##   step
```

```
library(irr)
```

```
## Loading required package: lpSolve
```

```

library(gt)

##
## Attaching package: 'gt'
##
## The following object is masked from 'package:cowplot':
##
##     as_gtable

#Plotting function helper

plot_normality <- function(data, column_name) {
  column_data <- data[[column_name]]
  data_clean <- na.omit(column_data)
  density_estimate <- density(data_clean)
  normal_max <- dnorm(mean(data_clean), mean = mean(data_clean), sd = sd(data_clean))
  ylim_max <- max(density_estimate$y, normal_max)
  hist(data_clean, main = NA, xlab = column_name, ylab = "Density",
        breaks = 20, col = "skyblue", border = "white", probability = TRUE,
        ylim = c(0, ylim_max * 1.1),
        las = 1)
  lines(density_estimate, col = "blue", lwd = 2, lty = 1)
  curve(dnorm(x, mean = mean(data_clean), sd = sd(data_clean)),
        col = "red", lwd = 2, lty = 2, add = TRUE)
  legend("topright", inset = c(0, -0.2), legend = c("Density", "Normal Distribution"), col = c("blue"
})

plot_violin <- function(data, column_name, group_column) {
  # Remove missing values and prepare the data
  data_clean <- data[!is.na(data[[column_name]]) & !is.na(data[[group_column]]), ]
  data_clean[[column_name]] <- scale(data_clean[[column_name]])
  # Create violin plot with grouping
  ggplot(data_clean, aes_string(x = group_column, y = column_name, fill = group_column)) +
    geom_violin(trim = FALSE, alpha = 0.7) +
    geom_boxplot(width = 0.1, position = position_dodge(0.9), alpha = 0.5 ) +
    scale_fill_manual(
      values = c("0" = "lightblue", "1" = "darkorange") # Custom colors
    ) +
    ylim(-4, 4) +
    scale_x_discrete(labels = c("0" = "NR", "1" = "R")) + # Relabel x-axis
    labs(x = NULL, y = column_name) + # Remove legend title
    theme_minimal() +
    theme(legend.position = "none") # Remove the legend
}

plot_violin_outlier_suppress <- function(data, column_name, group_column) {
  # Remove missing values and prepare the data
  data_clean <- data[!is.na(data[[column_name]]) & !is.na(data[[group_column]]), ]
  data_clean[[column_name]] <- scale(data_clean[[column_name]])
  # Create violin plot with grouping
  ggplot(data_clean, aes_string(x = group_column, y = column_name, fill = group_column)) +
    geom_violin(trim = FALSE, alpha = 0.7) +
    geom_boxplot(width = 0.1, position = position_dodge(0.9), alpha = 0.5, outlier.shape = NA ) +

```

```

    scale_fill_manual(
      values = c("0" = "lightblue", "1" = "darkorange") # Custom colors
    ) +
    ylim(-3.5, 3.5) +
    scale_x_discrete(labels = c("0" = "NR", "1" = "R")) + # Relabel x-axis
    labs(x = NULL, y = column_name) + # Remove legend title
    theme_minimal() +
    theme(legend.position = "none") # Remove the legend
  }
}

plot_outliers <- function(data, x_var, y_var, x_label, y_label, ymin, ymax, outliers) {
  outlier_CCIDs <- unique(outliers$CC.ID)
  data <- data %>%
    mutate(is_outlier = ifelse(CC.ID %in% outlier_CCIDs, TRUE, FALSE))

  ggplot(data, aes(x = .data[[x_var]], y = .data[[y_var]])) +
    geom_point(aes(color = is_outlier, alpha = is_outlier), size = 2, stroke = 1) +
    geom_line(aes(group = CC.ID), lty = 2, alpha = 0.2, color = "black") +
    geom_smooth(
      aes(group = CC.ID, color = is_outlier),
      method = "lm", se = FALSE, size = 0.5, lty = 1
    ) +
    labs(x = x_label, y = y_label) +
    ylim(ymin, ymax) +
    scale_color_manual(
      values = c("TRUE" = "green", "FALSE" = "darkorange"),
      labels = c("TRUE" = "Outlier", "FALSE" = "Non-Outlier")
    ) +
    scale_alpha_manual(values = c("TRUE" = 0.3, "FALSE" = 0.1), guide = "none") +
    theme_minimal() +
    theme(
      plot.title = element_text(hjust = 0.5),
      legend.position = "bottom",
      legend.box = "horizontal"
    ) +
    guides(color = guide_legend(title = NULL))
}

```

Mydata contains ALL cross-sectional and longitudinal data across all phases for 1) Cattell, 2) STW, 3) Memory Imm, 4) Memory Del, 5) Simple Spped (inv RT Simple), 6) Choice Speed (inv RT choice).

```

mydata <- read.csv("CambridgeCentreForAg-AllCognitiveData_DATA_2026-05-17_1254.csv")
phase3_dates <- read.csv("CambridgeCentreForAg-PHASE3DATES_DATA_2026-05-18_1537.csv")

```

#Data wrangling

```

mydata$phase <- ifelse(grepl("phase_1", mydata$redcap_event_name), "1",
  ifelse(grepl("phase_2", mydata$redcap_event_name), "2",
    ifelse(grepl("phase_3", mydata$redcap_event_name), "3",
      ifelse(grepl("phase_4", mydata$redcap_event_name), "4",
        ifelse(grepl("phase_5", mydata$redcap_event_name), "5",
          ifelse(grepl("participant_info", mydata$redcap_event_name), "999", NA))))))
mydata$arm <- ifelse(grepl("arm_1", mydata$redcap_event_name), "1",

```

```

      ifelse(grepl("arm_2", mydata$redcap_event_name), "2", NA))

names(mydata)[names(mydata) == "record_id"] <- "CC.ID"
names(mydata)[names(mydata) == "question_num_years_edu"] <- "education_years"
names(mydata)[names(mydata) == "phase_cog_age"] <- "Age"
names(mydata)[names(mydata) == "cattell_total_score"] <- "Cattell"
names(mydata)[names(mydata) == "stw_n_correct_trials"] <- "STW"
names(mydata)[names(mydata) == "story_memory_immediate_score"] <- "mem_imm"
names(mydata)[names(mydata) == "story_memory_delayed_score"] <- "mem_del"
names(mydata)[names(mydata) == "story_memory_recog_score"] <- "mem_recog"
names(mydata)[names(mydata) == "rt_simple_inv_rt_mean"] <- "simple_speed"
names(mydata)[names(mydata) == "rt_choice_inv_rt_mean"] <- "choice_speed"
names(mydata)[names(mydata) == "phase_cog_online_paper"] <- "format"

mydata$education_years <- as.numeric(mydata$education_years)

mydata <- mydata %>%
  group_by(CC.ID) %>%
  mutate(education_years = {
    val <- education_years[phase == 1 & !is.na(education_years)]
    if (length(val) > 0) val[1] else education_years
  }) %>%
  ungroup()

mydata <- mydata %>%
  group_by(CC.ID) %>%
  mutate(across(c(sex, dob, phase_code), ~ {
    val <- .x[phase == 999 & !is.na(.x)]
    if (length(val) > 0) val[1] else .x
  }))) %>%
  ungroup()

mydata <- mydata %>% filter(phase != 999 | is.na(phase))

cog_measures <- c("Cattell", "STW", "mem_imm", "mem_del",
  "simple_speed", "choice_speed")
mydata <- mydata %>% mutate(across(all_of(cog_measures), as.numeric))

mydata$wave <- NA
mydata$wave[mydata$phase == 1] <- 0
mydata$wave[mydata$phase == 2] <- 0
mydata$wave[mydata$phase == 4] <- 1
mydata$wave[mydata$phase == 5] <- 2

mydata$dob <- as.Date(mydata$dob)
mydata$phase_cog_date <- as.Date(mydata$phase_cog_date)
mydata$Age <- round(as.numeric((mydata$phase_cog_date - mydata$dob) / 365.25), 2)
mydata$sex <- as.factor(mydata$sex)

```

#REDCAP CORRECTION

```

mydata$mem_imm[mydata$CC.ID == 620697 & mydata$wave == 2] <- NA #REDCAP ENTRY MISTAKE
mydata$mem_del[mydata$CC.ID == 620697 & mydata$wave == 2] <- NA #REDCAP ENTRY MISTAKE

```

```

fix_ids <- c(220518, 310051, 420173, 520980, 620685)
fix_dates <- as.Date(c("2024-09-01", "2024-12-01", "2025-01-01", "2025-01-01", "2025-04-01")) #REDCAP NU
for (i in seq_along(fix_ids)) {
  mydata$phase_cog_date[mydata$CC.ID == fix_ids[i] & mydata$phase == "5"] <- fix_dates[i]
}
mydata$phase_cog_date[mydata$CC.ID == 420435 & mydata$phase == "5"] <- NA

phase3_dates$phase3_meg_date <- as.Date(phase3_dates$phase_meg_date)
phase3_dates$phase3_mri_date <- as.Date(phase3_dates$phase_mri_date)
phase3_dates$phase <- ifelse(grepl("phase_3", phase3_dates$redcap_event_name), "3", NA)
phase3_dates$CC.ID <- phase3_dates$record_id

mydata <- mydata %>% left_join(phase3_dates %>% select(CC.ID, phase3_meg_date, phase3_mri_date), by = "CC.ID")
mydata <- mydata %>% mutate(
  phase_cog_date = if_else(
    phase == "3",
    pmin(phase3_meg_date, phase3_mri_date, na.rm = TRUE),
    phase_cog_date)
)

```

```

subject_patterns <- mydata %>%
  mutate(
    participated = as.integer(if_any(all_of(cog_measures), ~ !is.na(.x)))
  ) %>%
  group_by(CC.ID) %>%
  summarize(
    wave0 = as.integer(any(participated == 1 & wave == 0)),
    wave1 = as.integer(any(participated == 1 & wave == 1)),
    wave2 = as.integer(any(participated == 1 & wave == 2)),
    .groups = "drop"
  ) %>%
  mutate(
    pattern = paste0(wave0, wave1, wave2),
    group = case_when(
      pattern == "100" ~ "G1",
      pattern == "110" ~ "G2",
      pattern %in% c("111", "101") ~ "G3",
      TRUE ~ "Other"
    )
  )

table(subject_patterns$pattern, useNA = "ifany")

```

```

##
## 000 010 100 101 110 111
## 35 3 2016 42 496 110

```

```

table(subject_patterns$group, useNA = "ifany")

```

```

##
## G1 G2 G3 Other
## 2016 496 152 38

```

```
mydata <- merge(mydata, subject_patterns, by = "CC.ID")

missing_ids <- read.csv("missing_record_ids.csv")
names(missing_ids)[names(missing_ids) == "record_id"] <- "CC.ID"
mydata <- mydata[!mydata$CC.ID %in% missing_ids$CC.ID, ]
mydata %>% group_by(group) %>% summarise(n_unique = n_distinct(CC.ID, na.rm = FALSE))
```

```
## # A tibble: 4 x 2
##   group n_unique
##   <chr>   <int>
## 1 G1      2003
## 2 G2      495
## 3 G3      152
## 4 Other    31
```

```
mydata$return_code <- ifelse(mydata$group == "G3", 1, 0)
mydata$return_code <- as.factor(mydata$return_code)
```

Attendance = non-NA phase_cog_date for that phase.
Each participant's set of attended phases drives which arrow they belong to.

```
att <- mydata %>%
  filter(!is.na(phase_cog_date), !is.na(phase)) %>%
  mutate(phase_label = ifelse(phase == "2" & !is.na(arm),
                              paste0("p2_arm", arm),
                              paste0("p", phase))) %>%
  distinct(CC.ID, phase_label) %>%
  mutate(present = 1L) %>%
  pivot_wider(names_from = phase_label, values_from = present, values_fill = 0L)

for (col in c("p1", "p2_arm1", "p2_arm2", "p3", "p4", "p5")) {
  if (!col %in% names(att)) att[[col]] <- 0L
}

arrow_counts <- tibble::tibble(
  arrow = c(
    "P1 -> P4 (skipping P2 & P3)",
    "P1 -> P2_arm2",
    "P2_arm2 -> P4",
    "P2_arm2 -> P5 (via P4)",
    "P2_arm2 -> P5 (skipping P4)",
    "P2_arm1 -> P4 (skipping P3)",
    "P2_arm1 -> P5 (via P4, skipping P3)",
    "P2_arm1 -> P5 (skipping P3 & P4)",
    "P3 -> P4",
    "P3 -> P5 (via P4)",
    "P3 -> P5 (skipping P4)"
  ),
  n = c(
    sum(att$p1 == 1 & att$p4 == 1 & att$p2_arm1 == 0 & att$p2_arm2 == 0 & att$p3 == 0),
    sum(att$p1 == 1 & att$p2_arm2 == 1),
    sum(att$p2_arm2 == 1 & att$p4 == 1),
    sum(att$p2_arm2 == 1 & att$p4 == 1 & att$p5 == 1),
  )
```

```

sum(att$p2_arm2 == 1 & att$p5 == 1 & att$p4 == 0),
sum(att$p2_arm1 == 1 & att$p4 == 1 & att$p3 == 0),
sum(att$p2_arm1 == 1 & att$p4 == 1 & att$p5 == 1 & att$p3 == 0),
sum(att$p2_arm1 == 1 & att$p5 == 1 & att$p3 == 0 & att$p4 == 0),
sum(att$p3 == 1 & att$p4 == 1),
sum(att$p3 == 1 & att$p4 == 1 & att$p5 == 1),
sum(att$p3 == 1 & att$p5 == 1 & att$p4 == 0)
)
)

totals <- tibble::tibble(
  phase = c("P1", "P2_arm1", "P2_arm2", "P3", "P4", "P5"),
  n      = c(sum(att$p1), sum(att$p2_arm1), sum(att$p2_arm2),
             sum(att$p3), sum(att$p4),      sum(att$p5))
)

cat("\nPhase totals (boxes on the diagram):\n"); print(totals)

```

```

##
## Phase totals (boxes on the diagram):

```

```

## # A tibble: 6 x 2
##   phase      n
##   <chr>  <int>
## 1 P1      2680
## 2 P2_arm1  681
## 3 P2_arm2   93
## 4 P3      278
## 5 P4      631
## 6 P5      152

```

```

cat("\nArrow counts (paths between phases):\n"); print(arrow_counts)

```

```

##
## Arrow counts (paths between phases):

```

```

## # A tibble: 11 x 2
##   arrow                                     n
##   <chr>                                <int>
## 1 P1 -> P4 (skipping P2 & P3)          345
## 2 P1 -> P2_arm2                        93
## 3 P2_arm2 -> P4                        42
## 4 P2_arm2 -> P5 (via P4)               17
## 5 P2_arm2 -> P5 (skipping P4)           3
## 6 P2_arm1 -> P4 (skipping P3)         115
## 7 P2_arm1 -> P5 (via P4, skipping P3)   39
## 8 P2_arm1 -> P5 (skipping P3 & P4)     17
## 9 P3 -> P4                            129
## 10 P3 -> P5 (via P4)                   56
## 11 P3 -> P5 (skipping P4)              20

```



```

trajectories <- att %>%
  mutate(path = paste0(
    ifelse(p1 == 1, "P1-", ""),
    ifelse(p2_arm1 == 1, "P2a1-", ""),
    ifelse(p2_arm2 == 1, "P2a2-", ""),
    ifelse(p3 == 1, "P3-", ""),
    ifelse(p4 == 1, "P4-", ""),
    ifelse(p5 == 1, "P5", "")
  )) %>%
  count(path, sort = TRUE)

diagram_table <- tibble::tibble(
  Section = c(rep("Phase totals (boxes)", 6),
    rep("Arrows (pathways between phases)", 11)),
  Label = c(
    "Phase 1 - CC3000 Home Interviews",
    "Phase 2 Arm 1 - CC700",
    "Phase 2 Arm 2 - CCfrail",
    "Phase 3 - CC280",
    "Phase 4 - CCenrich Cognitive Testing",
    "Phase 5 - CCrescan",
    "Phase 1 → Phase 4 (skipping P2 & P3)",
    "Phase 1 → Phase 2 Arm 2",
    "Phase 2 Arm 2 → Phase 4",
    "Phase 2 Arm 2 → Phase 5 (via P4)",
    "Phase 2 Arm 2 → Phase 5 (skipping P4)",
    "Phase 2 Arm 1 → Phase 4 (skipping P3)",
    "Phase 2 Arm 1 → Phase 5 (via P4, skipping P3)",
    "Phase 2 Arm 1 → Phase 5 (skipping P3 & P4)",
    "Phase 3 → Phase 4",
    "Phase 3 → Phase 5 (via P4)",
    "Phase 3 → Phase 5 (skipping P4)"
  ),
  n = c(totals$n, arrow_counts$n)
)

gt_tbl <- diagram_table %>%
  gt(groupname_col = "Section") %>%
  tab_header(
    title = md("**CamCAN participant flow**"),
    subtitle = "Phase totals and pathway counts"
  ) %>%
  cols_label(Label = "", n = "n") %>%
  cols_align(aligned = "right", columns = n) %>%
  tab_style(
    style = cell_text(weight = "bold"),
    locations = cells_row_groups()
  ) %>%
  tab_options(
    table.font.size = 13,
    heading.title.font.size = 16,
    row_group.font.weight = "bold"
  )

```

CamCAN participant flow

Phase totals and pathway counts

	n
Phase totals (boxes)	
Phase 1 — CC3000 Home Interviews	2680
Phase 2 Arm 1 — CC700	681
Phase 2 Arm 2 — CCfrail	93
Phase 3 — CC280	278
Phase 4 — CCenrich Cognitive Testing	631
Phase 5 — CCrescan	152
Arrows (pathways between phases)	
Phase 1 → Phase 4 (skipping P2 & P3)	345
Phase 1 → Phase 2 Arm 2	93
Phase 2 Arm 2 → Phase 4	42
Phase 2 Arm 2 → Phase 5 (via P4)	17
Phase 2 Arm 2 → Phase 5 (skipping P4)	3
Phase 2 Arm 1 → Phase 4 (skipping P3)	115
Phase 2 Arm 1 → Phase 5 (via P4, skipping P3)	39
Phase 2 Arm 1 → Phase 5 (skipping P3 & P4)	17
Phase 3 → Phase 4	129
Phase 3 → Phase 5 (via P4)	56
Phase 3 → Phase 5 (skipping P4)	20

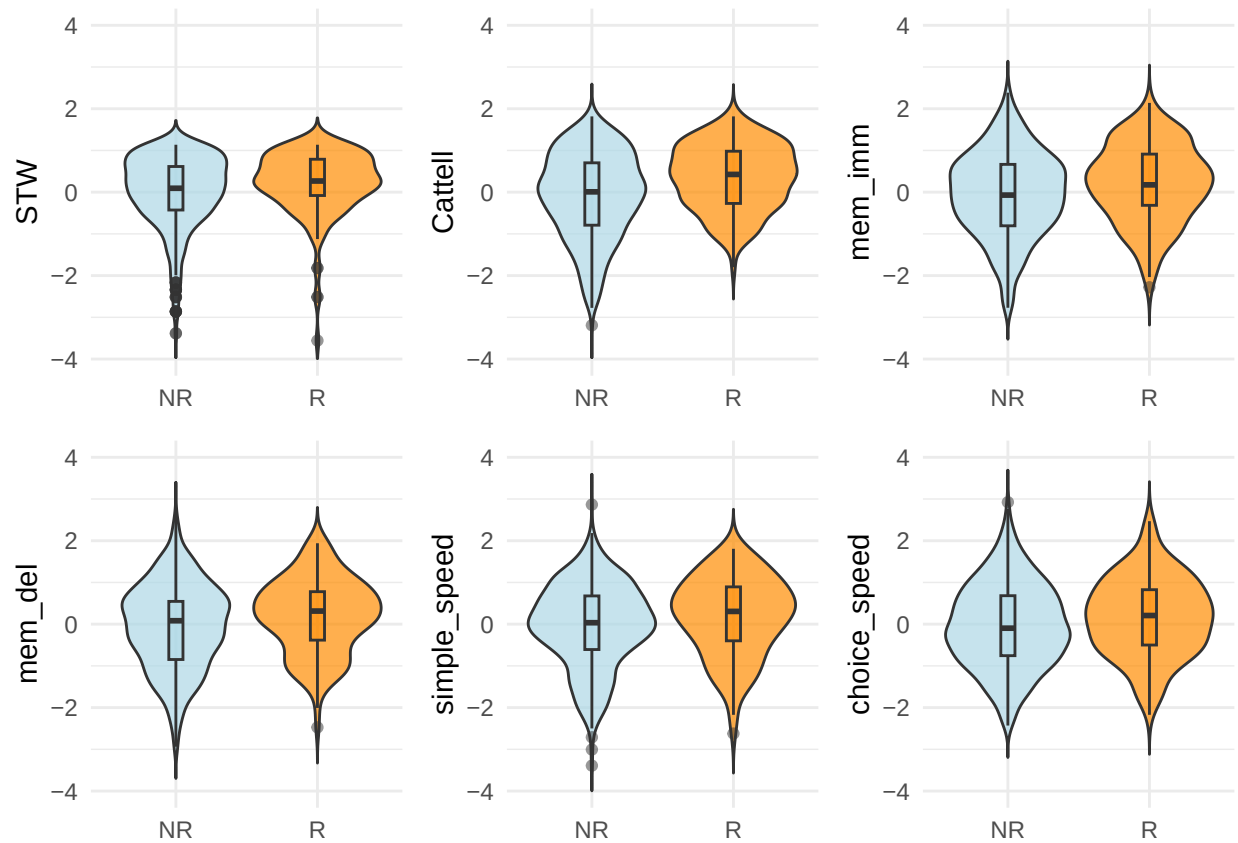
gt_tbl

```
CC700 <- read.csv("CC700_CC.ID.csv")
names(CC700)[names(CC700) == "record_id"] <- "CC.ID"
```

```
mydata700 <- mydata[mydata$CC.ID %in% CC700$CC.ID, ]
```

```
mydata1_R_NR <- mydata700[mydata700$phase == 1, ]
mydata2_R_NR <- mydata700[mydata700$phase == 2, ]
```

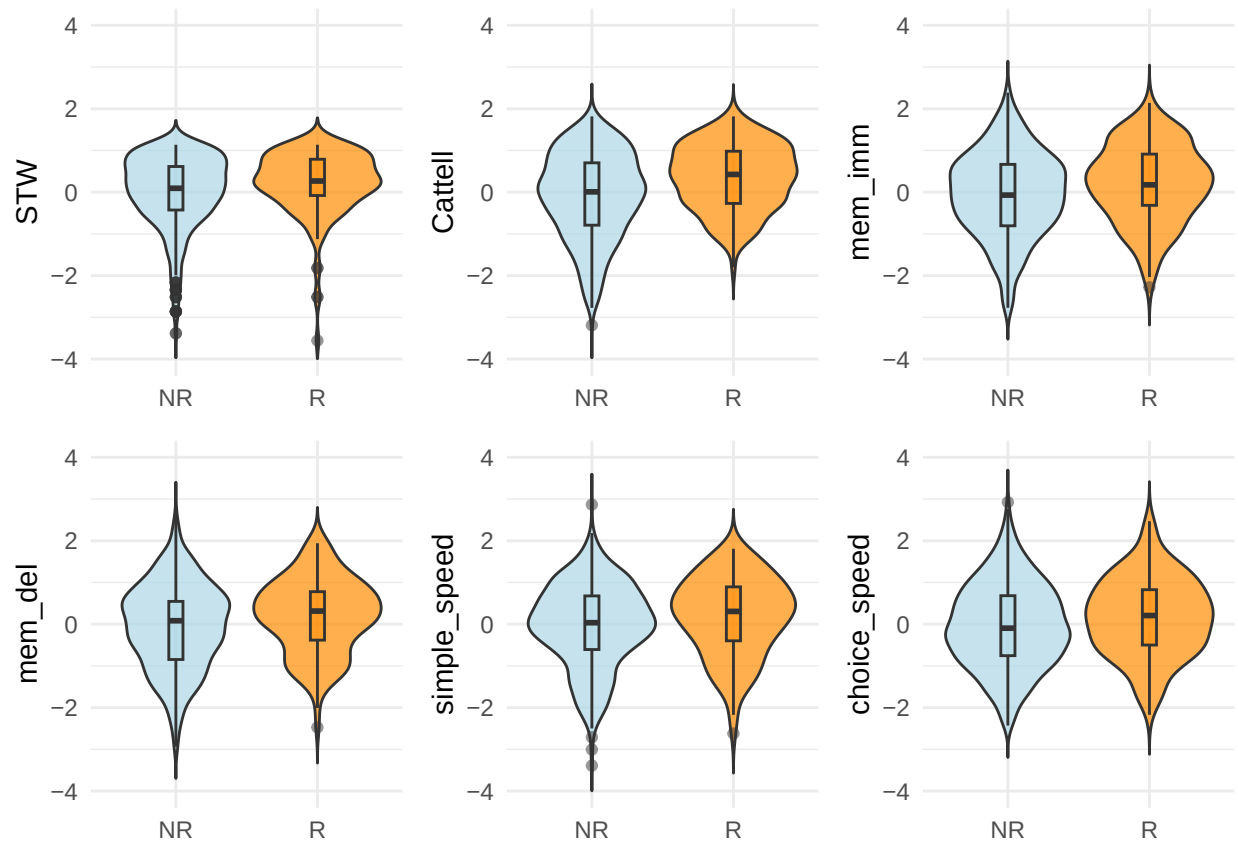
```
p1 <- plot_violin(mydata1_R_NR, "STW", "return_code")
p2 <- plot_violin(mydata2_R_NR, "Cattell", "return_code")
p3 <- plot_violin(mydata1_R_NR, "mem_imm", "return_code")
p4 <- plot_violin(mydata1_R_NR, "mem_del", "return_code")
p5 <- plot_violin(mydata1_R_NR, "simple_speed", "return_code")
p6 <- plot_violin(mydata1_R_NR, "choice_speed", "return_code")
grid.arrange(p1, p2, p3, p4, p5, p6, ncol = 3)
```



```
#Keep raw values for variables that will be transformed
mydata$STW_raw <- mydata$STW
mydata$STW      <- (mydata$STW_raw)^3

#check normality after transformation
p1 <- plot_violin(mydata1_R_NR, "STW", "return_code")
p2 <- plot_violin(mydata2_R_NR, "Cattell", "return_code")
p3 <- plot_violin(mydata1_R_NR, "mem_imm", "return_code")
p4 <- plot_violin(mydata1_R_NR, "mem_del", "return_code")
p5 <- plot_violin(mydata1_R_NR, "simple_speed", "return_code")
p6 <- plot_violin(mydata1_R_NR, "choice_speed", "return_code")

grid.arrange(p1, p2, p3, p4, p5, p6, ncol = 3)
```



Plot violins without outliers

```
# p1 <- plot_violin_outlier_suppress(mydata1_R_NR, "STW", "return_code")
# p2 <- plot_violin_outlier_suppress(mydata2_R_NR, "Cattell", "return_code")
# p3 <- plot_violin_outlier_suppress(mydata1_R_NR, "mem_imm", "return_code")
# p4 <- plot_violin_outlier_suppress(mydata1_R_NR, "mem_del", "return_code")
# p5 <- plot_violin_outlier_suppress(mydata1_R_NR, "simple_speed", "return_code")
# p6 <- plot_violin_outlier_suppress(mydata1_R_NR, "choice_speed", "return_code")

# grid.arrange(p1, p2, p3, p4, p5, p6, ncol = 3)
```

Phase separation between Phase 1 and Phase 2 is a practical helper due to Redcap construction. We treat cognitive variables collected in Phase 1 as Phase 2 interchangeably (due to a very short time lag).

```
mydata1_R <- mydata1_R_NR[mydata1_R_NR$group == "G3", ]
mydata2_R <- mydata2_R_NR[mydata2_R_NR$group == "G3", ]

mydata1_NR <- mydata1_R_NR[mydata1_R_NR$group != "G3" & !is.na(mydata1_R_NR$CC.ID), ]
mydata2_NR <- mydata2_R_NR[mydata2_R_NR$group != "G3" & !is.na(mydata2_R_NR$CC.ID), ]
```

AGE AND SEX DISTRIBUTION

```
summary(mydata$Age)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs  
##    18.49  43.92   64.17   61.11  79.17  102.17   6839
```

```
table(mydata$sex)           # 2 - female #1 - male
```

```
##  
##      1      2  
## 4871 6200
```

```
summary(mydata1_R$Age)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##    24.00  42.30   56.96   55.01  66.40   83.41
```

```
summary(mydata1_NR$Age)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NAs  
##    18.50  41.46   59.17   57.50  75.50   94.00      1
```

```
table(mydata1_R$sex)
```

```
##  
##      1      2  
## 89 63
```

```
table(mydata1_NR$sex)
```

```
##  
##      1      2  
## 299 325
```

```
sex_ratio <- table(mydata1_R_NR$return_code, mydata1_R_NR$sex)  
print(sex_ratio)
```

```
##  
##           1      2  
## 0 299 325  
## 1  89  63
```

```
chisq.test(sex_ratio)
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: sex_ratio
## X-squared = 5.1134, df = 1, p-value = 0.02374

m_age <- glm(return_code ~ Age, data = mydata1_R_NR, na.action = na.omit, family = binomial)
anova(m_age, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: return_code
##
## Terms added sequentially (first to last)
##
##
##      Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL                                774      767.23
## Age   1       2.16      773      765.07  0.1416
```

```
m_sex <- glm(return_code ~ sex, data = mydata1_R_NR, na.action = na.omit, family = binomial)
anova(m_sex, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: return_code
##
## Terms added sequentially (first to last)
##
##
##      Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL                                775      767.67
## sex   1     5.553      774      762.12  0.01845 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
m_edu <- glm(return_code ~ education_years, data = mydata1_R_NR, na.action = na.omit, family = binomial)
anova(m_edu, test = "Chisq")
```

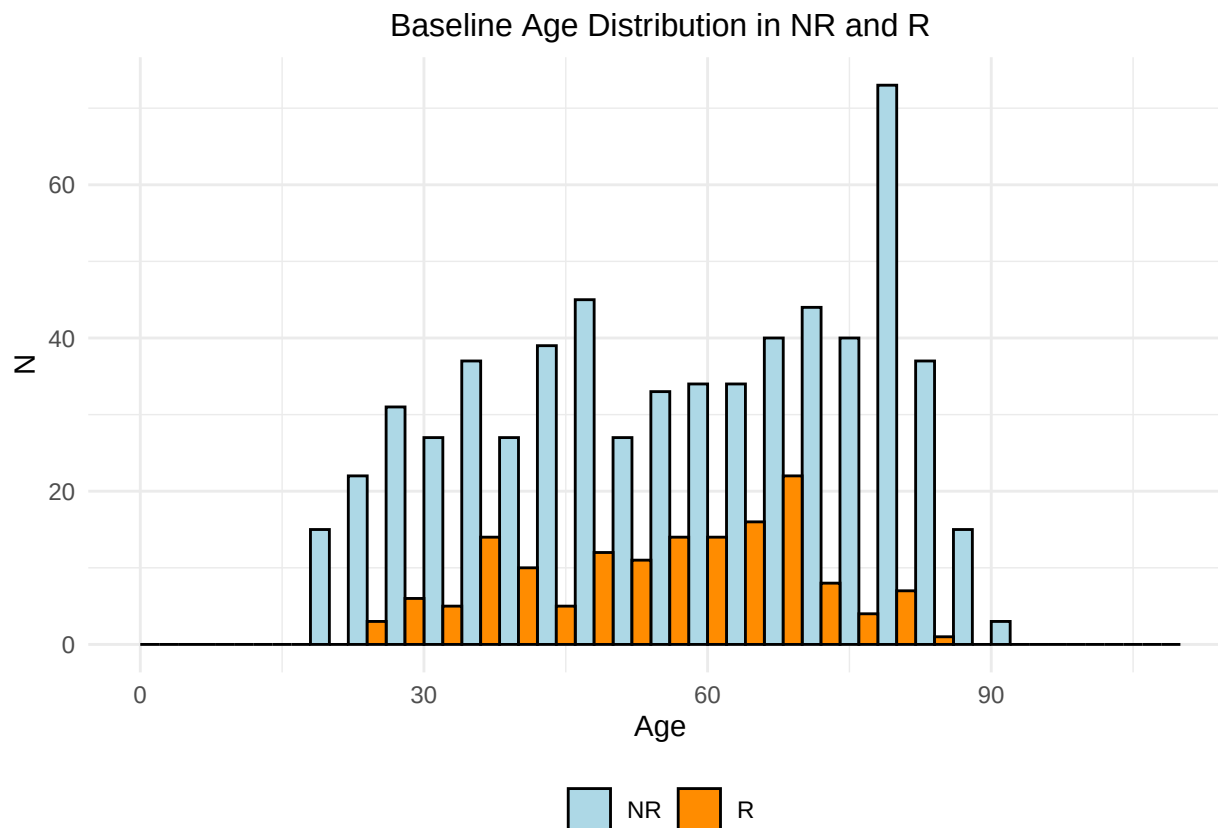
```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: return_code
##
## Terms added sequentially (first to last)
##
##
```

```
##              Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL              773      766.80
## education_years  1   21.511      772      745.28 3.518e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

m_full <- glm(return_code ~ sex*Age, data = mydata1_R_NR, na.action = na.omit, family = binomial)
anova(m_full, test = "Chisq")

## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: return_code
##
## Terms added sequentially (first to last)
##
##
##              Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL              774      767.23
## sex              1   5.6385      773      761.59 0.01757 *
## Age              1   2.3894      772      759.21 0.12216
## sex:Age          1   1.1725      771      758.03 0.27890
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

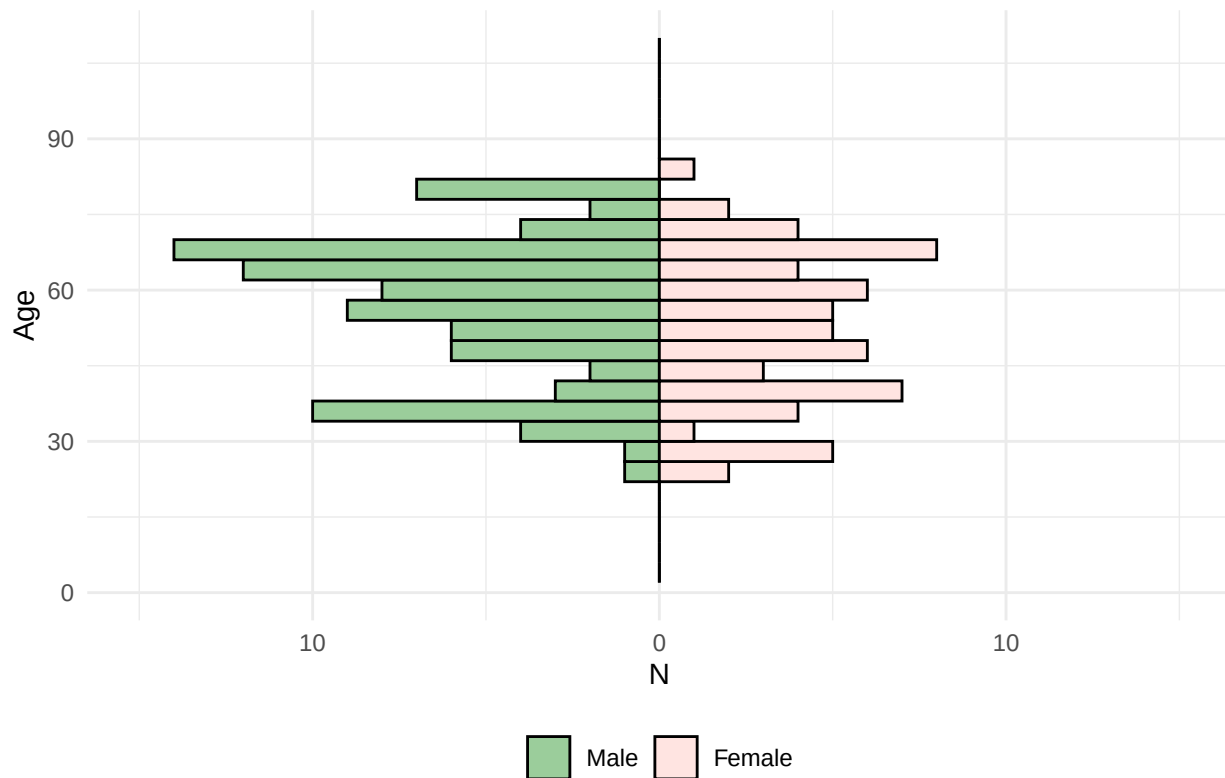
ggplot(mydata1_R_NR, aes(x = Age, fill = return_code)) +
  geom_histogram(binwidth = 4, position = "dodge", color = "black", size = 0.2) +
  labs(x = "Age", y = "N", fill = "Group") +
  ggtitle("Baseline Age Distribution in NR and R") +
  scale_fill_manual(values = c("lightblue", "darkorange"),
                    labels = c("0" = "NR", "1" = "R"), name = NULL) +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, size = 12),
        legend.position = "bottom") +
  xlim(0, 110)
```



```
mydata1_R <- mydata1_R %>%
  mutate(Age = as.numeric(Age),
         N = ifelse(sex == 1, -1, 1))

ggplot(mydata1_R, aes(x = Age, fill = factor(sex))) +
  geom_histogram(data = filter(mydata1_R, sex == 1), aes(y = -..count..),
                 binwidth = 4, color = "black", size = 0.2, position = "identity") +
  geom_histogram(data = filter(mydata1_R, sex == 2), aes(y = ..count..),
                 binwidth = 4, color = "black", size = 0.2, position = "identity") +
  scale_fill_manual(values = c("#9BCD9B", "#FFE4E1"),
                    labels = c("1" = "Male", "2" = "Female"), name = NULL) +
  labs(x = "Age", y = "N", fill = "sex") +
  ggtitle("Age Distribution by Sex in R at a Baseline") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, size = 14),
        legend.position = "bottom") +
  xlim(0, 110) +
  coord_flip() + # Flip coordinates for a horizontal layout
  scale_y_continuous(labels = abs, limits = c(-15, 15)) # Combine labels and limits
```


Age Distribution by Sex in R at a Baseline



EDUCATION

```

outliers_edu    <- boxplot.stats(mydata1_R_NR$education_years)$out
outliers_edu_R  <- boxplot.stats(mydata1_R$education_years)$out
outliers_edu_NR <- boxplot.stats(mydata1_NR$education_years)$out

mydata1_R_NR$education_years[mydata1_R_NR$return_code == 1 & mydata1_R_NR$education_years %in% outliers_edu]
mydata1_R_NR$education_years[mydata1_R_NR$return_code == 0 & mydata1_R_NR$education_years %in% outliers_edu_R]
mydata1_R_NR$education_years[mydata1_R_NR$return_code == 0 & mydata1_R_NR$education_years %in% outliers_edu_NR]

wilcox.test(education_years ~ return_code, data = mydata1_R_NR)

##
## Wilcoxon rank sum test with continuity correction
##
## data:  education_years by return_code
## W = 26917, p-value = 1.233e-10
## alternative hypothesis: true location shift is not equal to 0

summary(mydata1_R$Education)

## Length Class Mode
##      0  NULL  NULL
  
```

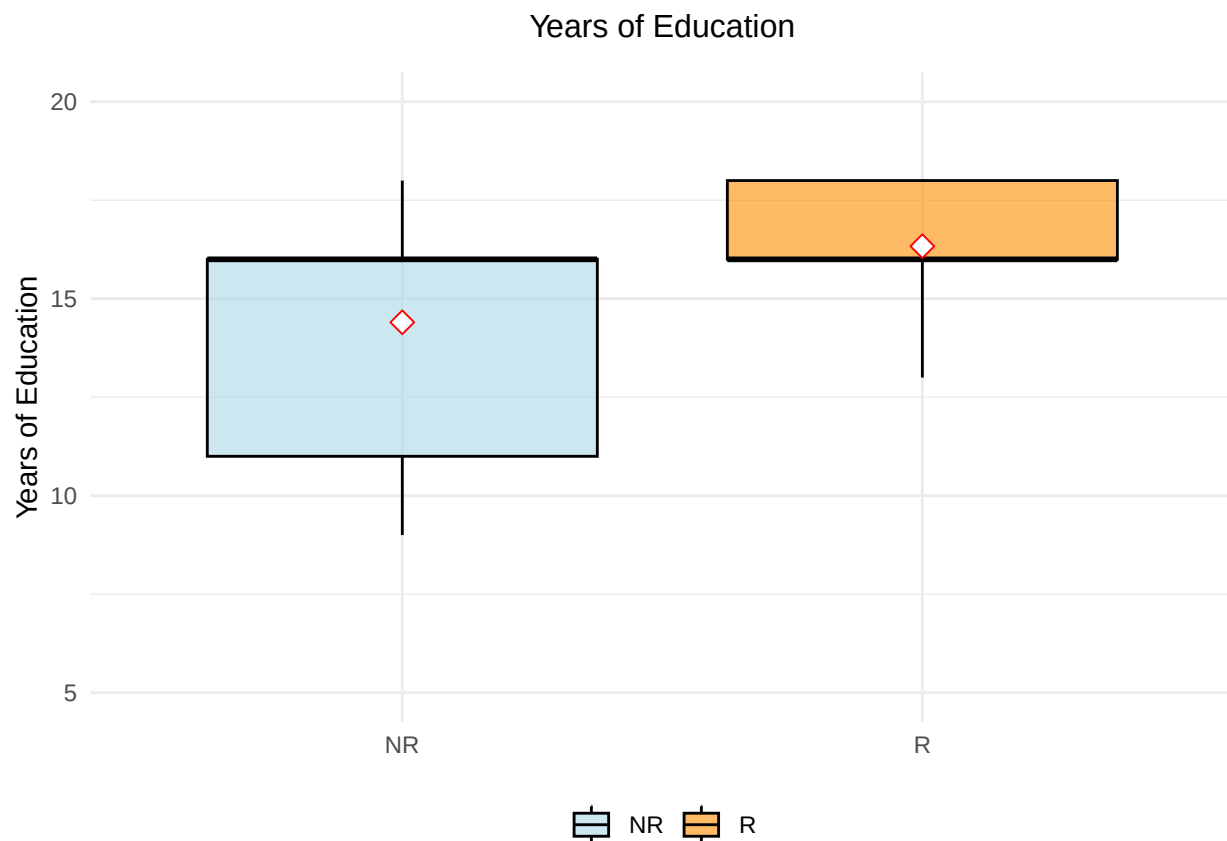
```
summary(mydata1_NR$Education)
```

```
## Length Class Mode  
##      0    NULL  NULL
```

```
ggplot(mydata1_R_NR, aes(x = factor(return_code), y = education_years, fill = factor(return_code))) +  
  geom_boxplot(color = "black", alpha = 0.6) + # Box plot with some transparency  
  stat_summary(fun = mean, geom = "point", shape = 23, size = 3, color = "red", fill = "white") +  
  labs(title = "Years of Education",  
        y = "Years of Education",  
        x = NULL) +  
  scale_fill_manual(values = c("0" = "lightblue", "1" = "darkorange"),  
                    labels = c("0" = "NR", "1" = "R"),  
                    name = NULL) +  
  scale_x_discrete(labels = c("0" = "NR", "1" = "R")) +  
  theme_minimal() +  
  theme(  
    legend.position = "bottom",  
    legend.background = element_rect(fill = "transparent", color = NA),  
    plot.title = element_text(hjust = 0.5, vjust = 3, size = 12)  
  ) +  
  ylim(5, 20)
```

```
## Warning: Removed 22 rows containing non-finite outside the scale range  
## ('stat_boxplot()').
```

```
## Warning: Removed 22 rows containing non-finite outside the scale range  
## ('stat_summary()').
```



Education selectivity

```
clean_data <- na.omit(mydata1_R_NR[, c("education_years", "Age", "return_code")])
education_model <- lm(education_years ~ poly(Age,2), data = clean_data)
clean_data$regressed_education <- residuals(education_model, na.rm = TRUE)

r_residuals_ed <- clean_data$regressed_education[clean_data$return_code == "1"]
cc700_residuals_ed <- clean_data$regressed_education

(mean(r_residuals_ed) - (mean(cc700_residuals_ed)))/sd(cc700_residuals_ed)
```

```
## [1] 0.4810646
```

```
(mean(mydata1_R$education_years, na.rm = TRUE) - (mean(mydata1_R_NR$education_years, na.rm = TRUE)))/sd
```

```
## [1] 0.2939932
```

```
clean_data <- na.omit(mydata1_R_NR[, c("education_years", "sex", "return_code", "Age")])
m_education <- lm(education_years ~ return_code * poly(Age, 2), data = clean_data)
summary(m_education)
```

```
##
```

```
## Call:
## lm(formula = education_years ~ return_code * poly(Age, 2), data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.8091 -2.1060  0.8379  1.8184  5.6228
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      14.4503     0.1079 133.933 < 2e-16 ***
## return_code1       1.8863     0.3068   6.148 1.27e-09 ***
## poly(Age, 2)1     -17.9054     2.8464  -6.291 5.38e-10 ***
## poly(Age, 2)2      -8.2758     2.8692  -2.884 0.00403 **
## return_code1:poly(Age, 2)1  13.2599     9.1450   1.450 0.14749
## return_code1:poly(Age, 2)2  10.2502     9.9133   1.034 0.30147
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.68 on 748 degrees of freedom
## Multiple R-squared:  0.1239, Adjusted R-squared:  0.118
## F-statistic: 21.15 on 5 and 748 DF, p-value: < 2.2e-16
```

```
car::Anova(m_education, type=3)
```

```
## Anova Table (Type III tests)
##
## Response: education_years
##              Sum Sq Df    F value    Pr(>F)
## (Intercept)    128815  1 17937.9821 < 2.2e-16 ***
## return_code       271  1   37.8014 1.274e-09 ***
## poly(Age, 2)     350  2   24.3747 5.557e-11 ***
## return_code:poly(Age, 2)  18  2    1.2389  0.2903
## Residuals       5371 748
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

FLUID INTELLIGENCE (CATTELL)

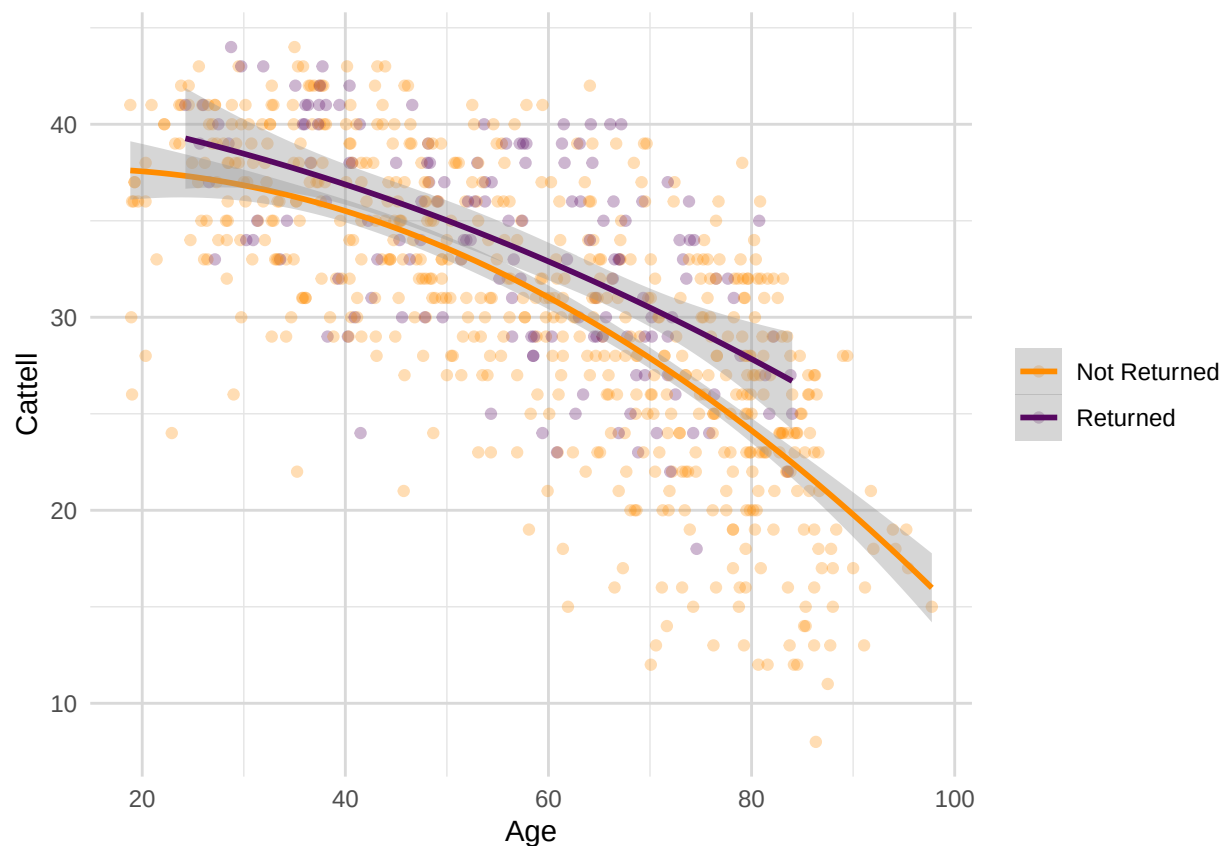
```
ggplot(mydata2_R_NR, aes(x = Age, y = Cattell, color = return_code)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), se = , size = 1) +
  labs(x = "Age", y = "Cattell") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14),
    plot.subtitle = element_text(hjust = 0.5, size = 12),
    panel.grid.major = element_line(color = "grey85", size = 0.5),
    panel.grid.minor = element_line(color = "grey90", size = 0.25)
  ) +
  scale_color_manual(values = c("darkorange", "#570861"), labels = c("0" = "Not Returned", "1" = "Returned"))
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

## Warning: The 'size' argument of 'element_line()' is deprecated as of ggplot2 3.4.0.
## i Please use the 'linewidth' argument instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

## Warning: Removed 29 rows containing non-finite outside the scale range
## ('stat_smooth()').

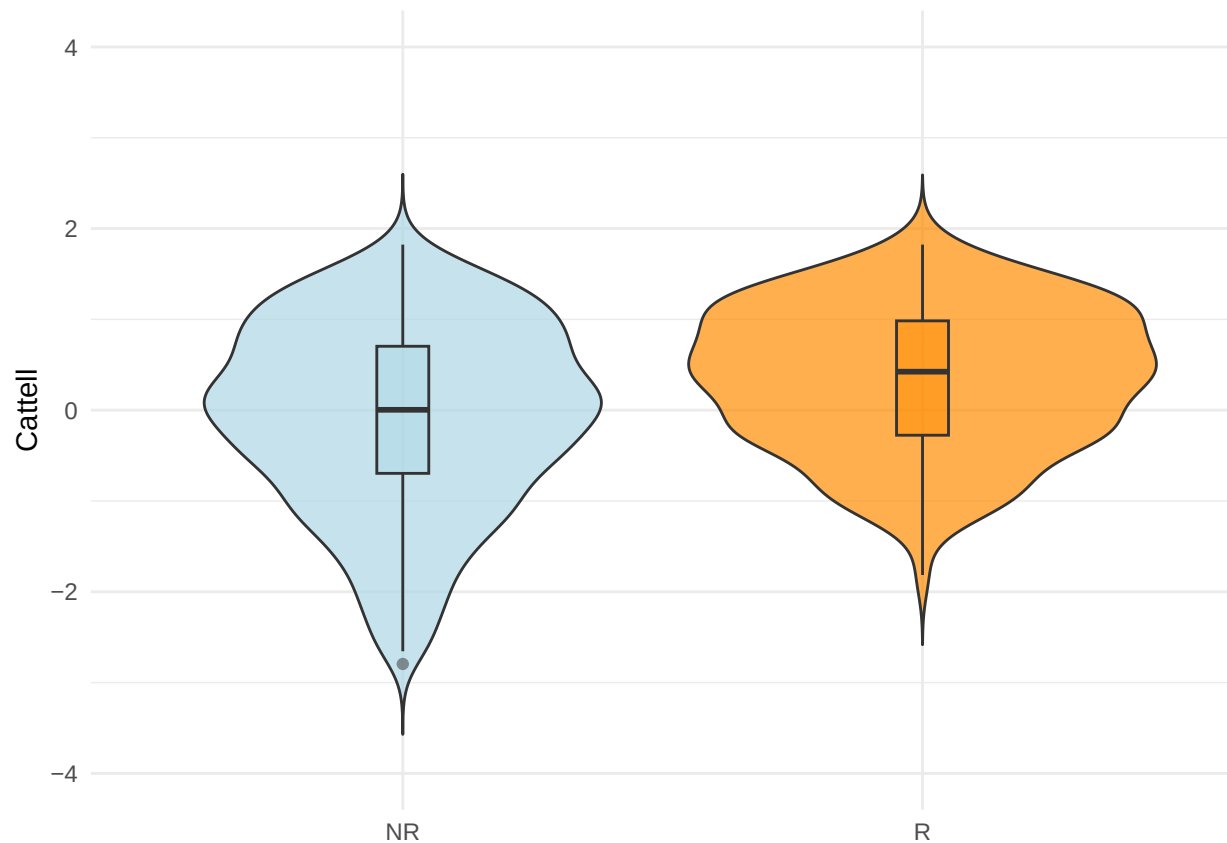
## Warning: Removed 29 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
outliers_cattell_R <- boxplot.stats(mydata2_R$Cattell)$out
outliers_cattell_NR <- boxplot.stats(mydata2_NR$Cattell)$out

mydata2_R_NR$Cattell[mydata2_R_NR$return_code == 1 & mydata2_R_NR$Cattell %in% outliers_cattell_R] <- NA
mydata2_R_NR$Cattell[mydata2_R_NR$return_code == 0 & mydata2_R_NR$Cattell %in% outliers_cattell_NR] <- NA

plot_violin(mydata2_R_NR, "Cattell", "return_code")
```



```
t.test(mydata2_R$Cattell, mydata2_NR$Cattell, paired = FALSE)
```

```
##
## Welch Two Sample t-test
##
## data: mydata2_R$Cattell and mydata2_NR$Cattell
## t = 5.9178, df = 303.83, p-value = 8.795e-09
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  2.141955 4.276105
## sample estimates:
## mean of x mean of y
##  33.50000 30.29097
```

```
wilcox.test(Cattell ~ return_code, data = mydata2_R_NR)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: Cattell by return_code
## W = 34450, p-value = 4.427e-06
## alternative hypothesis: true location shift is not equal to 0
```

Cattell Selectivity

```
clean_data <- na.omit(mydata2_R_NR[, c("Cattell", "Age", "education_years", "return_code", "sex")])
clean_data$return_code <- as.factor(clean_data$return_code)
clean_data$sex <- as.factor(clean_data$sex)

cat_model <- lm(Cattell ~ poly(Age,2), data = clean_data)
clean_data$regressed_cattell <- residuals(cat_model, na.rm = TRUE)

returners_residuals <- clean_data$regressed_cattell[clean_data$return_code == "1"]
cc700_residuals <- clean_data$regressed_cattell

#age-orthogonal selectivity statistic
(mean(returners_residuals) - mean(cc700_residuals)) / sd(cc700_residuals)

## [1] 0.3093427

#age-driven selectivity statistic
(mean(mydata2_R$Cattell, na.rm = TRUE) - (mean(mydata2_R_NR$Cattell, na.rm = TRUE)))/sd(mydata2_R_NR$Ca

## [1] 0.3537716

m_cattell <-lm(Cattell ~ return_code * scale(education_years) * poly(Age, 2) * sex, data = clean_data)
summary(m_cattell)

##
## Call:
## lm(formula = Cattell ~ return_code * scale(education_years) *
##      poly(Age, 2) * sex, data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.3818  -3.2177   0.3713   3.4910  11.1433
##
## Coefficients:
##                                     Estimate Std. Error
## (Intercept)                       31.1590     0.2859
## return_code1                       2.1418     0.7323
## scale(education_years)              2.3114     0.2930
## poly(Age, 2)1                    -122.6900     7.3811
## poly(Age, 2)2                     -25.7948     7.5740
## sex2                              -0.8162     0.4024
## return_code1:scale(education_years) -1.7056     0.8913
## return_code1:poly(Age, 2)1          4.6442    22.0633
## return_code1:poly(Age, 2)2         34.5040    22.8786
## scale(education_years):poly(Age, 2)1  5.7956     7.5658
## scale(education_years):poly(Age, 2)2 -6.7296     6.9926
## return_code1:sex2                   0.5353     1.2358
## scale(education_years):sex2         -0.8776     0.4168
## poly(Age, 2)1:sex2                  0.3259    10.5699
## poly(Age, 2)2:sex2                  5.5430    10.6322
```

```
## return_code1:scale(education_years):poly(Age, 2)1      15.0903      27.3630
## return_code1:scale(education_years):poly(Age, 2)2     -26.6348      28.1855
## return_code1:scale(education_years):sex2               1.3795       1.3120
## return_code1:poly(Age, 2)1:sex2                      -1.3990      36.6257
## return_code1:poly(Age, 2)2:sex2                       3.4391      40.3492
## scale(education_years):poly(Age, 2)1:sex2             0.5158      11.2743
## scale(education_years):poly(Age, 2)2:sex2             7.7569      10.7903
## return_code1:scale(education_years):poly(Age, 2)1:sex2 38.1164      41.5858
## return_code1:scale(education_years):poly(Age, 2)2:sex2  5.0286      42.3443
##
## t value Pr(>|t|)
## (Intercept)      108.988 < 2e-16 ***
## return_code1       2.925 0.003553 **
## scale(education_years) 7.889 1.13e-14 ***
## poly(Age, 2)1     -16.622 < 2e-16 ***
## poly(Age, 2)2      -3.406 0.000696 ***
## sex2              -2.028 0.042887 *
## return_code1:scale(education_years) -1.914 0.056069 .
## return_code1:poly(Age, 2)1      0.210 0.833340
## return_code1:poly(Age, 2)2      1.508 0.131958
## scale(education_years):poly(Age, 2)1      0.766 0.443914
## scale(education_years):poly(Age, 2)2     -0.962 0.336178
## return_code1:sex2      0.433 0.665038
## scale(education_years):sex2 -2.106 0.035576 *
## poly(Age, 2)1:sex2      0.031 0.975409
## poly(Age, 2)2:sex2      0.521 0.602292
## return_code1:scale(education_years):poly(Age, 2)1      0.551 0.581471
## return_code1:scale(education_years):poly(Age, 2)2     -0.945 0.344986
## return_code1:scale(education_years):sex2      1.051 0.293409
## return_code1:poly(Age, 2)1:sex2 -0.038 0.969540
## return_code1:poly(Age, 2)2:sex2      0.085 0.932099
## scale(education_years):poly(Age, 2)1:sex2      0.046 0.963521
## scale(education_years):poly(Age, 2)2:sex2      0.719 0.472451
## return_code1:scale(education_years):poly(Age, 2)1:sex2  0.917 0.359672
## return_code1:scale(education_years):poly(Age, 2)2:sex2  0.119 0.905503
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.706 on 721 degrees of freedom
## Multiple R-squared:  0.5815, Adjusted R-squared:  0.5681
## F-statistic: 43.55 on 23 and 721 DF, p-value: < 2.2e-16
```

```
car::Anova(m_cattell, type=3)
```

```
## Anova Table (Type III tests)
##
## Response: Cattell
##
## Sum Sq Df F value
## (Intercept) 263063 1 11878.4722
## return_code 189 1 8.5553
## scale(education_years) 1378 1 62.2417
## poly(Age, 2) 6451 2 145.6478
## sex 91 1 4.1144
## return_code:scale(education_years) 81 1 3.6618
## return_code:poly(Age, 2) 53 2 1.1910
```



```
## scale(education_years):poly(Age, 2)          29  2    0.6454
## return_code:sex                             4  1    0.1876
## scale(education_years):sex                  98  1    4.4340
## poly(Age, 2):sex                            6  2    0.1362
## return_code:scale(education_years):poly(Age, 2) 25  2    0.5716
## return_code:scale(education_years):sex         24  1    1.1055
## return_code:poly(Age, 2):sex                 0  2    0.0045
## scale(education_years):poly(Age, 2):sex       12  2    0.2765
## return_code:scale(education_years):poly(Age, 2):sex 21  2    0.4639
## Residuals                                15967 721
##                                           Pr(>F)
## (Intercept)                            < 2.2e-16 ***
## return_code                           0.003553 **
## scale(education_years)                1.128e-14 ***
## poly(Age, 2)                          < 2.2e-16 ***
## sex                                    0.042887 *
## return_code:scale(education_years)     0.056069 .
## return_code:poly(Age, 2)              0.304501
## scale(education_years):poly(Age, 2)    0.524757
## return_code:sex                       0.665038
## scale(education_years):sex             0.035576 *
## poly(Age, 2):sex                      0.872671
## return_code:scale(education_years):poly(Age, 2) 0.564876
## return_code:scale(education_years):sex 0.293409
## return_code:poly(Age, 2):sex          0.995545
## scale(education_years):poly(Age, 2):sex 0.758497
## return_code:scale(education_years):poly(Age, 2):sex 0.629014
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Cattell Attrition (Pattern-mixture) & ICC

```
df_cattell <- mydata %>%
  filter(group %in% c("G2", "G3"),
         phase %in% c(2, 4, 5)) %>%
  group_by(CC.ID) %>%
  filter(sum(!is.na(Cattell)) >= 2) %>%
  ungroup()

df_cattell <- df_cattell %>%
  group_by(CC.ID) %>%
  mutate(A0 = as.numeric(Age[wave == '0']),
         dA = as.numeric(Age - A0))

df_cattell$prac <- 1
df_cattell$prac[str_detect(df_cattell$wave, "0")] <- 0
df_cattell$prac <- factor(df_cattell$prac)
df_cattell$format <- factor(df_cattell$format)
df_cattell$format <- ifelse(df_cattell$wave == 0 & is.na(df_cattell$format), 1, df_cattell$format)

CC.ID <- df_cattell$CC.ID[df_cattell$wave == 0]
```

```

C0    <- df_cattell$Cattell[df_cattell$wave == 0]
C1    <- df_cattell$Cattell[df_cattell$wave == 1]
C2    <- df_cattell$Cattell[df_cattell$wave == 2]
educ  <- df_cattell$education_years[df_cattell$wave == 0]
sex    <- df_cattell$sex[df_cattell$wave == 0]
A0     <- df_cattell$Age[df_cattell$wave == 0]

feature_matrix <- data.frame(CC.ID, C0, C1, C2, A0, sex, educ)

set.seed(987)
iso <- isolation.forest(feature_matrix[, c("C0", "C1", "C2", "A0")], ntrees = 100, sample_size = 128, n

feature_matrix$pred <- predict(iso, feature_matrix[, c("C0", "C1", "C2", "A0")])
outliers <- feature_matrix[feature_matrix$pred > 0.60, c("CC.ID", "C0", "C1", "C2", "A0", "pred")]
outliers_cattell <- outliers

plot_outliers(df_cattell, "Age", "Cattell", "Age", "Cattell", min(df_cattell$Cattell), max(df_cattell$C

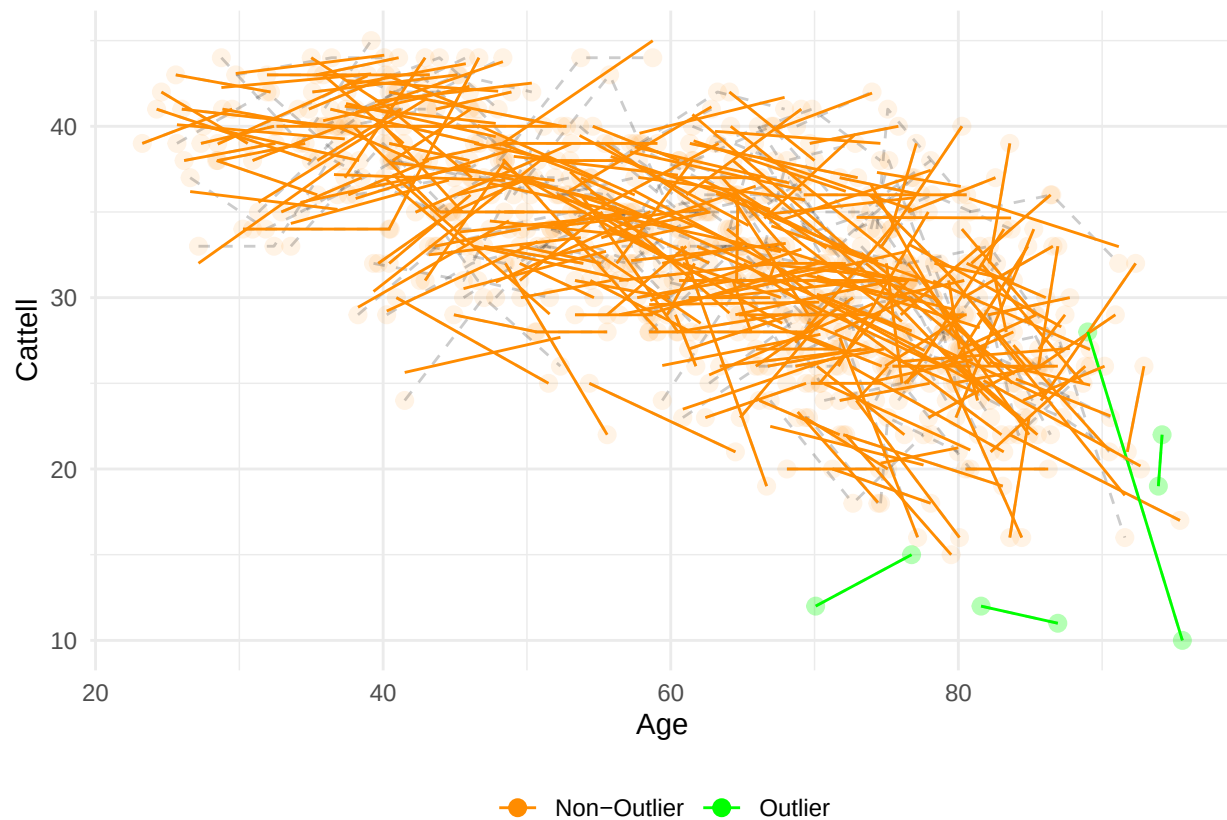
## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 207 rows containing non-finite outside the scale range
## ('stat_smooth()').

## Warning: Removed 207 rows containing missing values or values outside the scale range
## ('geom_point()').

## Warning: Removed 203 rows containing missing values or values outside the scale range
## ('geom_line()').

```



```
df_cattell$cattell[df_cattell$CC.ID %in% outliers_cattell$CC.ID] <- NA
```

```
## Warning: Unknown or uninitialised column: 'cattell'.
```

```
df_cattell$cattell_scaled[df_cattell$CC.ID %in% outliers_cattell$CC.ID] <- NA
```

```
## Warning: Unknown or uninitialised column: 'cattell_scaled'.
```

```
m_cattell = lmer('scale(Cattell) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + f
summary(m_cattell)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## "scale(Cattell) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + format + prac +
## Data: df_cattell
##
## REML criterion at convergence: 1337.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.6146 -0.4903  0.0270  0.5123  2.7748
##
## Random effects:
```

```
## Groups      Name      Variance Std.Dev.
## CC.ID      (Intercept) 0.3097   0.5565
## Residual                0.1792   0.4233
## Number of obs: 711, groups: CC.ID, 306
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)    -0.69016    0.16729  617.79306  -4.126
## scale(A0)       -0.81453    0.09030  655.16421  -9.020
## scale(dA)       -0.21625    0.06808  454.37324  -3.176
## groupG3          0.13899    0.07800  355.99697   1.782
## scale(education_years) 0.29087    0.06059  298.55232   4.800
## sex2            -0.10114    0.07811  299.75128  -1.295
## format          0.25644    0.07060  515.74744   3.632
## prac1           0.39417    0.09034  451.75313   4.363
## scale(A0):scale(dA)  -0.10053    0.06765  470.51275  -1.486
## scale(A0):groupG3    0.09196    0.07837  347.88164   1.173
## scale(dA):groupG3    0.05690    0.05030  436.19619   1.131
## scale(A0):scale(education_years) 0.08523    0.06474  304.14026   1.316
## scale(A0):sex2       0.17346    0.07901  302.22272   2.196
## scale(education_years):sex2 -0.10985    0.08774  300.87113  -1.252
## scale(A0):prac1      0.05668    0.08387  461.69207   0.676
## scale(A0):scale(dA):groupG3 -0.02693    0.04875  446.06757  -0.552
## scale(A0):scale(education_years):sex2 -0.06036    0.08798  304.31224  -0.686
##
##              Pr(>|t|)
## (Intercept)    4.20e-05 ***
## scale(A0)       < 2e-16 ***
## scale(dA)       0.001592 **
## groupG3         0.075608 .
## scale(education_years) 2.51e-06 ***
## sex2            0.196352
## format          0.000309 ***
## prac1           1.59e-05 ***
## scale(A0):scale(dA) 0.137972
## scale(A0):groupG3  0.241461
## scale(dA):groupG3  0.258584
## scale(A0):scale(education_years) 0.189055
## scale(A0):sex2     0.028886 *
## scale(education_years):sex2 0.211531
## scale(A0):prac1    0.499508
## scale(A0):scale(dA):groupG3 0.580936
## scale(A0):scale(education_years):sex2 0.493184
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)      if you need it
```

```
wide <- pivot_wider(df_cattell[df_cattell$group == "G3" & df_cattell$wave %in% c(0, 2)],
  id_cols = CC.ID,
  names_from = wave,
  values_from = Cattell)
```

```
mat <- wide[, -1]
icc(mat, model = "twoway", type = "consistency", unit = "single")
```

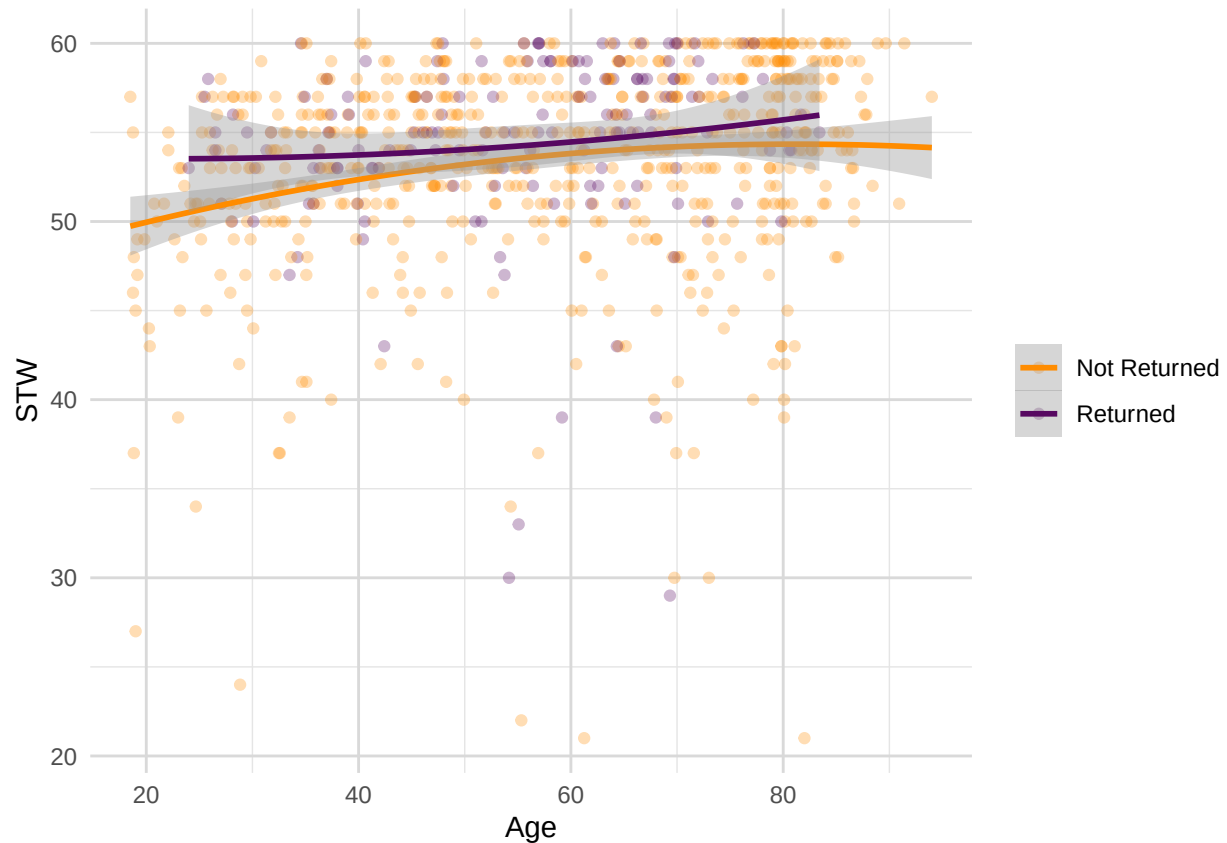
```
## Single Score Intraclass Correlation
##
## Model: twoway
## Type : consistency
##
## Subjects = 148
## Raters = 2
## ICC(C,1) = 0.813
##
## F-Test, H0: r0 = 0 ; H1: r0 > 0
## F(147,147) = 9.67 , p = 1.4e-36
##
## 95%-Confidence Interval for ICC Population Values:
## 0.75 < ICC < 0.861
```

STW

```
ggplot(mydata1_R_NR, aes(x = Age, y = STW, color = return_code)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), se = , size = 1) +
  labs(x = "Age", y = "STW") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14),
    plot.subtitle = element_text(hjust = 0.5, size = 12),
    panel.grid.major = element_line(color = "grey85", size = 0.5),
    panel.grid.minor = element_line(color = "grey90", size = 0.25)
  ) +
  scale_color_manual(values = c("darkorange", "#570861"), labels = c("0" = "Not Returned", "1" = "Returned"),
    name = NULL
  )
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
sum(is.na(mydata1_R_NR$STW) == TRUE) #N = 2 missing
```

```
## [1] 2
```

```
sum(is.na(mydata1_R$STW) == TRUE) #N = 0 missing
```

```
## [1] 0
```

```
sum(is.na(mydata1_NR$STW) == TRUE) #N = 2 missing
```

```
## [1] 2
```

```
outliers_stw_R <- boxplot.stats(mydata1_R$STW)$out
outliers_stw_NR <- boxplot.stats(mydata1_NR$STW)$out
```

```
mydata1_R_NR$STW[mydata1_R_NR$return_code == 1 & mydata1_R_NR$STW %in% outliers_stw_R] <- NA
mydata1_R_NR$STW[mydata1_R_NR$return_code == 0 & mydata1_R_NR$STW %in% outliers_stw_NR] <- NA
```

```
mydata1_R <- mydata1_R_NR[mydata1_R_NR$phase == 1 & mydata1_R_NR$return_code == 1, ]
mydata1_NR <- mydata1_R_NR[mydata1_R_NR$phase == 1 & mydata1_R_NR$return_code == 0, ]
```

```
sum(!is.na(mydata1_R_NR$STW) == TRUE)
```

```
## [1] 739
```

```
sum(!is.na(mydata1_R$STW) == TRUE)
```

```
## [1] 145
```

```
sum(!is.na(mydata1_NR$STW) == TRUE)
```

```
## [1] 594
```

```
t.test(mydata1_R$STW, mydata1_NR$STW, paired = FALSE)
```

```
##  
## Welch Two Sample t-test  
##  
## data: mydata1_R$STW and mydata1_NR$STW  
## t = 3.555, df = 275.77, p-value = 0.0004445  
## alternative hypothesis: true difference in means is not equal to 0  
## 95 percent confidence interval:  
## 0.5103212 1.7768261  
## sample estimates:  
## mean of x mean of y  
## 55.23448 54.09091
```

```
wilcox.test(STW ~ return_code, data = mydata1_R_NR)
```

```
##  
## Wilcoxon rank sum test with continuity correction  
##  
## data: STW by return_code  
## W = 37227, p-value = 0.01104  
## alternative hypothesis: true location shift is not equal to 0
```

STW Selectivity

```
clean_data <- na.omit(mydata1_R_NR[, c("STW", "Age", "education_years", "return_code", "sex")])  
clean_data$return_code <- as.factor(clean_data$return_code)  
clean_data$sex <- as.factor(clean_data$sex)  
  
stw_model <- lm(STW ~ poly(Age,2), data = clean_data)  
clean_data$regressed_stw <- residuals(stw_model, na.rm = TRUE)  
  
returners_residuals <- clean_data$regressed_stw[clean_data$return_code == "1"]  
cc700_residuals <- clean_data$regressed_stw  
  
(mean(returners_residuals) - mean(cc700_residuals)) / sd(cc700_residuals)
```

```
## [1] 0.311835
```

```
(mean(mydata1_R$STW, na.rm = TRUE) - (mean(mydata1_R_NR$STW, na.rm = TRUE)))/sd(mydata1_R_NR$STW, na.rm
```

```
## [1] 0.2249765
```

```
m_stw <-lm(STW ~ return_code*poly(Age,2)*scale(education_years)*sex , data = clean_data)
summary(m_stw)
```

```
##
## Call:
## lm(formula = STW ~ return_code * poly(Age, 2) * scale(education_years) *
##     sex, data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.5275  -2.0469   0.3487   2.3522   8.3554
##
## Coefficients:
##                                     Estimate Std. Error
## (Intercept)                       54.50134    0.21110
## return_code1                       0.70381    0.78619
## poly(Age, 2)1                      27.93620    5.41218
## poly(Age, 2)2                       3.93726    5.63742
## scale(education_years)              2.01658    0.21643
## sex2                               -0.35427    0.29440
## return_code1:poly(Age, 2)1          0.53033    24.18971
## return_code1:poly(Age, 2)2        -17.12334    23.58383
## return_code1:scale(education_years) -1.42445    1.15857
## poly(Age, 2)1:scale(education_years) -0.49354    5.60256
## poly(Age, 2)2:scale(education_years)  2.86639    5.20826
## return_code1:sex2                   1.44265    1.36156
## poly(Age, 2)1:sex2                 21.49057    7.63932
## poly(Age, 2)2:sex2                -11.11974    7.72740
## scale(education_years):sex2         -0.09764    0.30257
## return_code1:poly(Age, 2)1:scale(education_years)  4.16587    35.46805
## return_code1:poly(Age, 2)2:scale(education_years) -30.27757    35.78720
## return_code1:poly(Age, 2)1:sex2      5.82732    43.36247
## return_code1:poly(Age, 2)2:sex2     52.55800    45.79383
## return_code1:scale(education_years):sex2 -0.07067    2.04189
## poly(Age, 2)1:scale(education_years):sex2  9.31354    8.15921
## poly(Age, 2)2:scale(education_years):sex2 -7.25977    7.68040
## return_code1:poly(Age, 2)1:scale(education_years):sex2  7.93795    66.57079
## return_code1:poly(Age, 2)2:scale(education_years):sex2 35.39475    65.53125
##                                     t value Pr(>|t|)
## (Intercept)                     258.173 < 2e-16 ***
## return_code1                      0.895  0.37098
## poly(Age, 2)1                     5.162 3.19e-07 ***
## poly(Age, 2)2                     0.698  0.48515
## scale(education_years)            9.318 < 2e-16 ***
## sex2                             -1.203  0.22924
## return_code1:poly(Age, 2)1         0.022  0.98251
## return_code1:poly(Age, 2)2        -0.726  0.46804
## return_code1:scale(education_years) -1.229  0.21930
```



```
## poly(Age, 2)1:scale(education_years) -0.088 0.92983
## poly(Age, 2)2:scale(education_years) 0.550 0.58225
## return_code1:sex2 1.060 0.28971
## poly(Age, 2)1:sex2 2.813 0.00504 **
## poly(Age, 2)2:sex2 -1.439 0.15060
## scale(education_years):sex2 -0.323 0.74703
## return_code1:poly(Age, 2)1:scale(education_years) 0.117 0.90653
## return_code1:poly(Age, 2)2:scale(education_years) -0.846 0.39782
## return_code1:poly(Age, 2)1:sex2 0.134 0.89314
## return_code1:poly(Age, 2)2:sex2 1.148 0.25148
## return_code1:scale(education_years):sex2 -0.035 0.97240
## poly(Age, 2)1:scale(education_years):sex2 1.141 0.25406
## poly(Age, 2)2:scale(education_years):sex2 -0.945 0.34487
## return_code1:poly(Age, 2)1:scale(education_years):sex2 0.119 0.90512
## return_code1:poly(Age, 2)2:scale(education_years):sex2 0.540 0.58929
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.456 on 698 degrees of freedom
## Multiple R-squared: 0.3129, Adjusted R-squared: 0.2903
## F-statistic: 13.82 on 23 and 698 DF, p-value: < 2.2e-16
```

```
car::Anova(m_stw, type=3)
```

```
## Anova Table (Type III tests)
##
## Response: STW
##
## Sum Sq Df F value
## (Intercept) 796155 1 66653.3928
## return_code 10 1 0.8014
## poly(Age, 2) 328 2 13.7180
## scale(education_years) 1037 1 86.8167
## sex 17 1 1.4481
## return_code:poly(Age, 2) 7 2 0.2789
## return_code:scale(education_years) 18 1 1.5117
## poly(Age, 2):scale(education_years) 4 2 0.1519
## return_code:sex 13 1 1.1227
## poly(Age, 2):sex 120 2 5.0045
## scale(education_years):sex 1 1 0.1041
## return_code:poly(Age, 2):scale(education_years) 9 2 0.3586
## return_code:poly(Age, 2):sex 16 2 0.6857
## return_code:scale(education_years):sex 0 1 0.0012
## poly(Age, 2):scale(education_years):sex 22 2 0.9047
## return_code:poly(Age, 2):scale(education_years):sex 4 2 0.1529
## Residuals 8337 698
## Pr(>F)
## (Intercept) < 2.2e-16 ***
## return_code 0.37098
## poly(Age, 2) 1.434e-06 ***
## scale(education_years) < 2.2e-16 ***
## sex 0.22924
## return_code:poly(Age, 2) 0.75673
## return_code:scale(education_years) 0.21930
## poly(Age, 2):scale(education_years) 0.85913
```

```
## return_code:sex 0.28971
## poly(Age, 2):sex 0.00695 **
## scale(education_years):sex 0.74703
## return_code:poly(Age, 2):scale(education_years) 0.69877
## return_code:poly(Age, 2):sex 0.50408
## return_code:scale(education_years):sex 0.97240
## poly(Age, 2):scale(education_years):sex 0.40512
## return_code:poly(Age, 2):scale(education_years):sex 0.85822
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

STW Attrition and ICC

```
df_stw <- mydata %>%
  filter(group %in% c("G2", "G3"),
         phase %in% c(1, 4, 5)) %>%
  group_by(CC.ID) %>%
  filter(sum(!is.na(STW)) >= 2) %>%
  ungroup()

df_stw <- df_stw %>%
  group_by(CC.ID) %>%
  mutate(A0 = as.numeric(Age[wave == '0']),
         dA = as.numeric(Age - A0))

df_stw$prac <- 1
df_stw$prac[str_detect(df_stw$wave, "0")] <- 0
df_stw$prac <- factor(df_stw$prac)
df_stw$format <- factor(df_stw$format)
df_stw$format <- ifelse(df_stw$wave == 0 & is.na(df_stw$format), 1, df_stw$format)

CC.ID <- df_stw$CC.ID[df_stw$wave == 0]
C0 <- df_stw$STW[df_stw$wave == 0]
C1 <- df_stw$STW[df_stw$wave == 1]
C2 <- df_stw$STW[df_stw$wave == 2]
educ <- df_stw$education_years[df_stw$wave == 0]
sex <- df_stw$sex[df_stw$wave == 0]
A0 <- df_stw$Age[df_stw$wave == 0]

feature_matrix <- data.frame(CC.ID, C0, C1, C2, A0, sex, educ)

set.seed(987)
iso <- isolation.forest(feature_matrix[, c("C0", "C1", "C2", "A0")], ntrees = 100, sample_size = 128, n

feature_matrix$pred <- predict(iso, feature_matrix[, c("C0", "C1", "C2", "A0")])
outliers <- feature_matrix[feature_matrix$pred > 0.60, c("CC.ID", "C0", "C1", "C2", "A0", "pred")]
outliers_stw <- outliers

plot_outliers(df_stw, "Age", "STW", "Age", "STW", min(df_stw$STW), max(df_stw$STW), outliers = outliers,

## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 524 rows containing non-finite outside the scale range
## ('stat_smooth()').

## Warning: Removed 524 rows containing missing values or values outside the scale range
## ('geom_point()').

## Warning: Removed 520 rows containing missing values or values outside the scale range
## ('geom_line()').
```



```
df_stw$STW[df_stw$CC.ID %in% outliers_stw$CC.ID] <- NA
```

```
m_STW = lmer('scale(STW) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + format + j
summary(m_STW)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## "scale(STW) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + format + prac + sca
## Data: df_stw
##
## REML criterion at convergence: 3048
##
## Scaled residuals:
## Min 1Q Median 3Q Max
```

```

## -6.2100 -0.4163 0.0704 0.5040 2.9892
##
## Random effects:
## Groups Name Variance Std.Dev.
## CC.ID (Intercept) 0.4236 0.6508
## Residual 0.3052 0.5525
## Number of obs: 1308, groups: CC.ID, 606
##
## Fixed effects:
## Estimate Std. Error df t value
## (Intercept) 1.234e-01 1.156e-01 1.131e+03 1.067
## scale(A0) -3.769e-03 1.087e-01 1.170e+03 -0.035
## scale(dA) -1.777e-03 8.594e-02 7.869e+02 -0.021
## groupG3 8.531e-02 7.958e-02 6.905e+02 1.072
## scale(education_years) 4.191e-01 5.630e-02 5.851e+02 7.444
## sex2 -3.826e-02 6.552e-02 5.915e+02 -0.584
## format -1.241e-01 6.526e-02 9.746e+02 -1.902
## prac1 9.557e-02 1.389e-01 8.081e+02 0.688
## scale(A0):scale(dA) -1.413e-01 9.610e-02 7.997e+02 -1.471
## scale(A0):groupG3 1.014e-01 8.691e-02 6.868e+02 1.166
## scale(dA):groupG3 -2.751e-02 5.018e-02 7.576e+02 -0.548
## scale(A0):scale(education_years) 1.558e-01 6.210e-02 5.953e+02 2.509
## scale(A0):sex2 3.281e-01 6.698e-02 5.976e+02 4.899
## scale(education_years):sex2 4.078e-02 7.314e-02 5.881e+02 0.558
## scale(A0):prac1 2.043e-01 1.492e-01 8.023e+02 1.369
## scale(A0):scale(dA):groupG3 8.916e-02 5.359e-02 7.603e+02 1.664
## scale(A0):scale(education_years):sex2 -1.135e-01 7.746e-02 5.976e+02 -1.466
## Pr(>|t|)
## (Intercept) 0.2860
## scale(A0) 0.9724
## scale(dA) 0.9835
## groupG3 0.2841
## scale(education_years) 3.50e-13 ***
## sex2 0.5595
## format 0.0575 .
## prac1 0.4915
## scale(A0):scale(dA) 0.1418
## scale(A0):groupG3 0.2439
## scale(dA):groupG3 0.5836
## scale(A0):scale(education_years) 0.0124 *
## scale(A0):sex2 1.24e-06 ***
## scale(education_years):sex2 0.5774
## scale(A0):prac1 0.1713
## scale(A0):scale(dA):groupG3 0.0966 .
## scale(A0):scale(education_years):sex2 0.1432
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE) or
## vcov(x) if you need it

```

```
wide <- pivot_wider(df_stw[df_stw$group == "G3" & df_stw$wave %in% c(0, 2)],
  id_cols = CC.ID,
  names_from = wave,
  values_from = STW)
mat <- wide[, -1]
icc(mat, model = "twoway", type = "consistency", unit = "single")
```

```
## Single Score Intraclass Correlation
##
## Model: twoway
## Type : consistency
##
## Subjects = 143
## Raters = 2
## ICC(C,1) = 0.551
##
## F-Test, H0: r0 = 0 ; H1: r0 > 0
## F(142,142) = 3.45 , p = 4.32e-13
##
## 95%-Confidence Interval for ICC Population Values:
## 0.425 < ICC < 0.655
```

```
#sensitivity check
df_sub <- df_stw[df_stw$group == "G3", ]
mod <- lmer(STW ~ 1 + (1 | CC.ID), data = df_sub)
vc <- as.data.frame(VarCorr(mod))
ICC_C <- vc$vcov[vc$grp == "CC.ID"] /
  (vc$vcov[vc$grp == "CC.ID"] + vc$vcov[vc$grp == "Residual"])
ICC_C
```

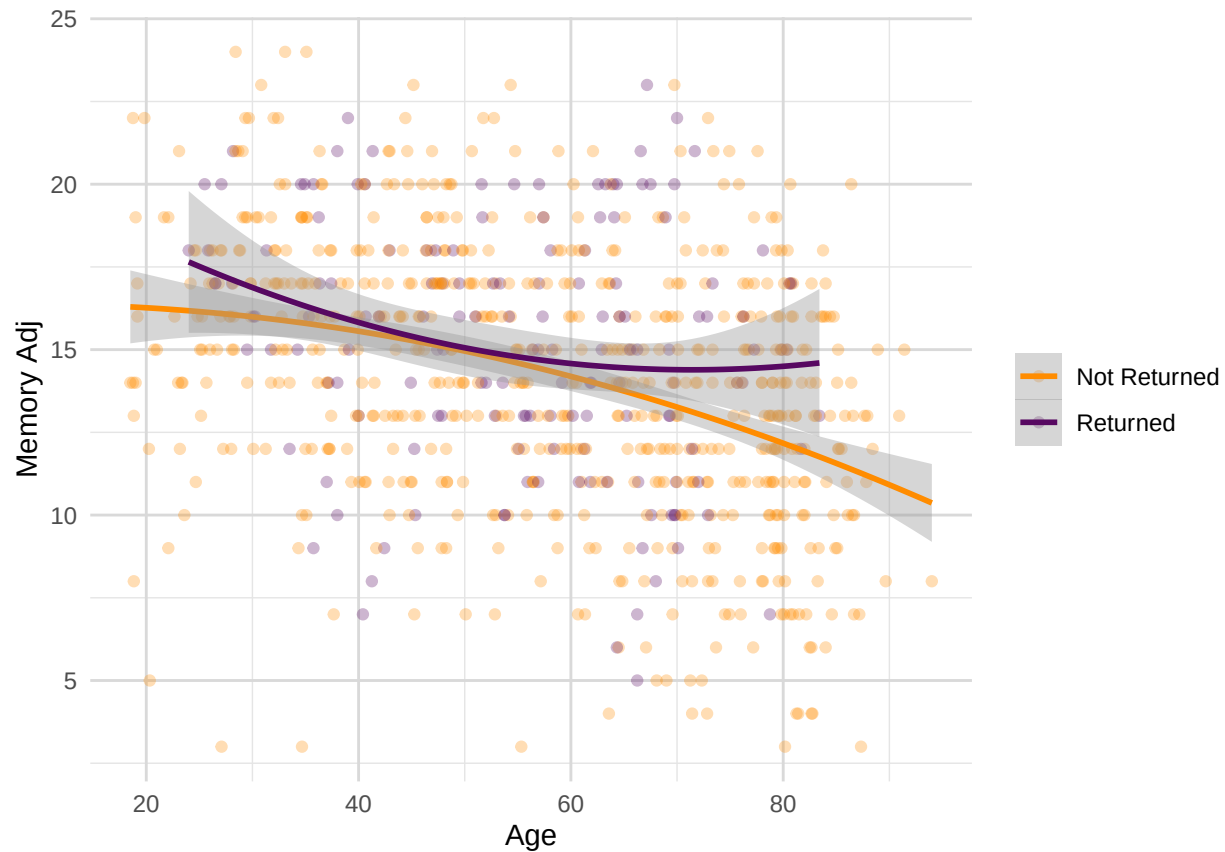
```
## [1] 0.6423334
```

VERBAL EPISODIC MEMORY

```
ggplot(mydata1_R_NR, aes(x = Age, y = mem_imm, color = return_code)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), se = TRUE , size = 1) +
  labs(x = "Age", y = "Memory Adj") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14),
    plot.subtitle = element_text(hjust = 0.5, size = 12),
    panel.grid.major = element_line(color = "grey85", size = 0.5),
    panel.grid.minor = element_line(color = "grey90", size = 0.25)
  ) +
  scale_color_manual(values = c("darkorange", "#570861"), labels = c("0" = "Not Returned", "1" = "Returned"),
    name = NULL
  )
```

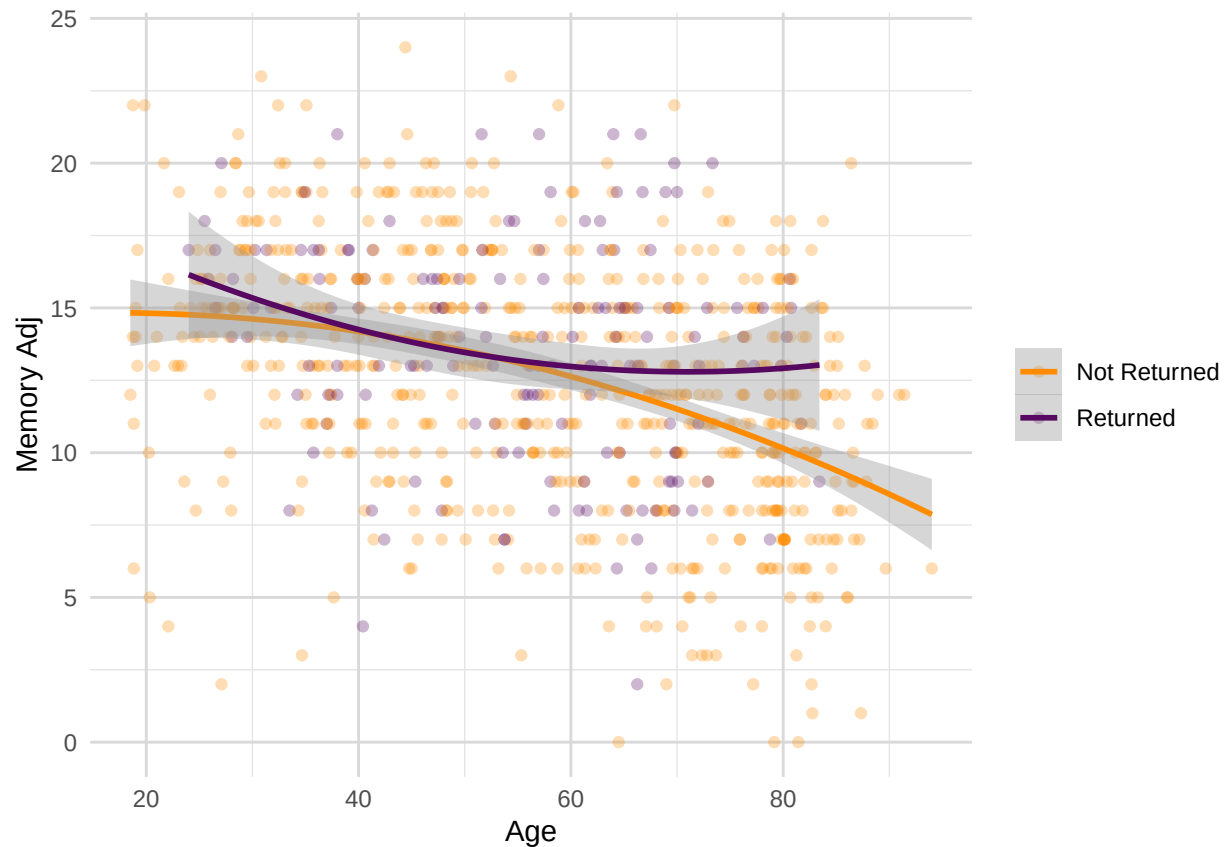
```
## Warning: Removed 1 row containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```
ggplot(mydata1_R_NR, aes(x = Age, y = mem_del, color = return_code)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), se = TRUE, size = 1) +
  labs(x = "Age", y = "Memory Adj") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14),
    plot.subtitle = element_text(hjust = 0.5, size = 12),
    panel.grid.major = element_line(color = "grey85", size = 0.5),
    panel.grid.minor = element_line(color = "grey90", size = 0.25)
  ) +
  scale_color_manual(values = c("darkorange", "#570861"), labels = c("0" = "Not Returned", "1" = "Returned"),
    name = NULL
  )
```

```
## Warning: Removed 1 row containing non-finite outside the scale range ('stat_smooth()').
## Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



```

outliers_mem_imm_R <- boxplot.stats(mydata1_R$mem_imm)$out
outliers_mem_imm_NR <- boxplot.stats(mydata1_NR$mem_imm)$out

mydata1_R_NR$mem_imm[mydata1_R_NR$return_code == 1 & mydata1_R_NR$mem_imm %in% outliers_mem_imm_R] <- NA
mydata1_R_NR$mem_imm[mydata1_R_NR$return_code == 0 & mydata1_R_NR$mem_imm %in% outliers_mem_imm_NR] <- NA

outliers_mem_del_R <- boxplot.stats(mydata1_R$mem_del)$out
outliers_mem_del_NR <- boxplot.stats(mydata1_NR$mem_del)$out

mydata1_R_NR$mem_del[mydata1_R_NR$return_code == 1 & mydata1_R_NR$mem_del %in% outliers_mem_del_R] <- NA
mydata1_R_NR$mem_del[mydata1_R_NR$return_code == 0 & mydata1_R_NR$mem_del %in% outliers_mem_del_NR] <- NA

wilcox.test(mem_imm ~ return_code, data = mydata1_R_NR)

```

```

##
## Wilcoxon rank sum test with continuity correction
##
## data: mem_imm by return_code
## W = 40134, p-value = 0.004983
## alternative hypothesis: true location shift is not equal to 0

```

```

wilcox.test(mem_del ~ return_code, data = mydata1_R_NR)

```

```

##

```

```
## Wilcoxon rank sum test with continuity correction
##
## data: mem_del by return_code
## W = 40080, p-value = 0.004659
## alternative hypothesis: true location shift is not equal to 0
```

Memory Selectivity

```
#selectivity age-driven
(mean(mydata1_R$mem_imm, na.rm = TRUE) - (mean(mydata1_R_NR$mem_imm, na.rm = TRUE)))/sd(mydata1_R_NR$mem_imm)
```

```
## [1] 0.1973288
```

```
(mean(mydata1_R$mem_del, na.rm = TRUE) - (mean(mydata1_R_NR$mem_del, na.rm = TRUE)))/sd(mydata1_R_NR$mem_del)
```

```
## [1] 0.1984596
```

```
# selectivity age-orthogonal (and education-adjusted) for Imm
clean_data <- na.omit(mydata1_R_NR[, c("mem_imm", "Age", "education_years", "return_code")])
imm_model <- lm(mem_imm ~ poly(Age,2), data = clean_data)
clean_data$regressed_imm <- residuals(imm_model, na.rm = TRUE)

returners_residuais <- clean_data$regressed_imm[clean_data$return_code == "1"]
cc700_residuais <- clean_data$regressed_imm
(mean(returners_residuais) - mean(cc700_residuais)) / sd(cc700_residuais)
```

```
## [1] 0.1965595
```

```
m_mem <- lm(mem_imm ~ return_code*poly(Age,2)*scale(education_years) , data = clean_data)
summary(m_mem)
```

```
##
## Call:
## lm(formula = mem_imm ~ return_code * poly(Age, 2) * scale(education_years),
##     data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.3716  -2.3997   0.2003   2.4571  11.1752
##
## Coefficients:
##              Estimate Std. Error t value
## (Intercept)      14.1976     0.1540  92.184
## return_code1         1.0513     0.6698   1.570
## poly(Age, 2)1    -34.8789     4.0745  -8.560
## poly(Age, 2)2     -5.2299     4.1437  -1.262
## scale(education_years)  0.9281     0.1566   5.926
## return_code1:poly(Age, 2)1  40.6262    20.1991   2.011
## return_code1:poly(Age, 2)2  34.7553    21.2138   1.638
```



```
## return_code1:scale(education_years)          -0.5794      0.9197  -0.630
## poly(Age, 2)1:scale(education_years)         -6.3192      4.2803  -1.476
## poly(Age, 2)2:scale(education_years)         -1.1259      4.0609  -0.277
## return_code1:poly(Age, 2)1:scale(education_years) -32.7095     28.4053  -1.152
## return_code1:poly(Age, 2)2:scale(education_years) -34.4891     29.4549  -1.171
##                                     Pr(>|t|)
## (Intercept)                                < 2e-16 ***
## return_code1                                0.1169
## poly(Age, 2)1                                < 2e-16 ***
## poly(Age, 2)2                                0.2073
## scale(education_years)                     4.76e-09 ***
## return_code1:poly(Age, 2)1                   0.0447 *
## return_code1:poly(Age, 2)2                   0.1018
## return_code1:scale(education_years)          0.5289
## poly(Age, 2)1:scale(education_years)         0.1403
## poly(Age, 2)2:scale(education_years)         0.7817
## return_code1:poly(Age, 2)1:scale(education_years) 0.2499
## return_code1:poly(Age, 2)2:scale(education_years) 0.2420
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.716 on 742 degrees of freedom
## Multiple R-squared:  0.1807, Adjusted R-squared:  0.1685
## F-statistic: 14.88 on 11 and 742 DF,  p-value: < 2.2e-16
```

```
car::Anova(m_mem, type=3, statistic = F)
```

```
## Anova Table (Type III tests)
##
## Response: mem_imm
##                                     Sum Sq Df  F value    Pr(>F)
## (Intercept)                     117321   1 8497.9053 < 2.2e-16
## return_code                        34     1   2.4637  0.11693
## poly(Age, 2)                     1033    2   37.4178 3.295e-16
## scale(education_years)            485     1   35.1143 4.759e-09
## return_code:poly(Age, 2)           94     2    3.4221 0.03316
## return_code:scale(education_years)   5     1    0.3969 0.52889
## poly(Age, 2):scale(education_years)  34     2    1.2386 0.29039
## return_code:poly(Age, 2):scale(education_years)  33    2    1.1835 0.30680
## Residuals                       10244  742
##
## (Intercept)                        ***
## return_code
## poly(Age, 2)                        ***
## scale(education_years)              ***
## return_code:poly(Age, 2)            *
## return_code:scale(education_years)
## poly(Age, 2):scale(education_years)
## return_code:poly(Age, 2):scale(education_years)
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```

# selectivity age-orthogonal (and education-adjusted) for Del
clean_data <- na.omit(mydata1_R_NR[, c("mem_del", "Age", "education_years", "return_code")])
del_model <- lm(mem_del ~ poly(Age,2), data = clean_data)
clean_data$regressed_del <- residuals(del_model, na.rm = TRUE)

returners_residuals <- clean_data$regressed_del[clean_data$return_code == "1"]
cc700_residuals <- clean_data$regressed_del
(mean(returners_residuals) - mean(cc700_residuals)) / sd(cc700_residuals)

```

```
## [1] 0.1914703
```

```

m_mem <-lm(mem_del ~ return_code*poly(Age,2)*scale(education_years) , data = clean_data)
summary(m_mem)

```

```

##
## Call:
## lm(formula = mem_del ~ return_code * poly(Age, 2) * scale(education_years),
##     data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.7106  -2.3883   0.2294   2.5909  12.0307
##
## Coefficients:
##                                     Estimate Std. Error t value
## (Intercept)                      12.5571     0.1603  78.339
## return_code1                       0.9964     0.6971   1.429
## poly(Age, 2)1                    -41.7200     4.2406  -9.838
## poly(Age, 2)2                      -9.1790     4.3126  -2.128
## scale(education_years)              0.9167     0.1630   5.624
## return_code1:poly(Age, 2)1          46.0306    21.0225   2.190
## return_code1:poly(Age, 2)2          46.4661    22.0786   2.105
## return_code1:scale(education_years) -0.2610     0.9572  -0.273
## poly(Age, 2)1:scale(education_years) -5.8245     4.4548  -1.307
## poly(Age, 2)2:scale(education_years) -4.3988     4.2264  -1.041
## return_code1:poly(Age, 2)1:scale(education_years) -31.3588    29.5632  -1.061
## return_code1:poly(Age, 2)2:scale(education_years) -44.9730    30.6556  -1.467
##                                     Pr(>|t|)
## (Intercept) < 2e-16 ***
## return_code1 0.1533
## poly(Age, 2)1 < 2e-16 ***
## poly(Age, 2)2 0.0336 *
## scale(education_years) 2.65e-08 ***
## return_code1:poly(Age, 2)1 0.0289 *
## return_code1:poly(Age, 2)2 0.0357 *
## return_code1:scale(education_years) 0.7852
## poly(Age, 2)1:scale(education_years) 0.1915
## poly(Age, 2)2:scale(education_years) 0.2983
## return_code1:poly(Age, 2)1:scale(education_years) 0.2892
## return_code1:poly(Age, 2)2:scale(education_years) 0.1428
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
##
## Residual standard error: 3.867 on 742 degrees of freedom
## Multiple R-squared:  0.2087, Adjusted R-squared:  0.1969
## F-statistic: 17.79 on 11 and 742 DF,  p-value: < 2.2e-16
```

```
car::Anova(m_mem, type=3, statistic = F)
```

```
## Anova Table (Type III tests)
##
## Response: mem_del
##
##              Sum Sq Df  F value    Pr(>F)
## (Intercept)    91774  1 6136.9587 < 2.2e-16
## return_code      31   1    2.0434  0.153287
## poly(Age, 2)    1514   2   50.6260 < 2.2e-16
## scale(education_years)  473  1   31.6269 2.647e-08
## return_code:poly(Age, 2)  140  2    4.6921  0.009441
## return_code:scale(education_years)  1  1    0.0743  0.785212
## poly(Age, 2):scale(education_years)  50  2    1.6880  0.185598
## return_code:poly(Age, 2):scale(education_years)  43  2    1.4498  0.235280
## Residuals      11096 742
##
## (Intercept)          ***
## return_code
## poly(Age, 2)          ***
## scale(education_years) ***
## return_code:poly(Age, 2) **
## return_code:scale(education_years)
## poly(Age, 2):scale(education_years)
## return_code:poly(Age, 2):scale(education_years)
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Memory ICC

```
df_mem <- mydata %>%
  filter(group %in% c("G2", "G3"),
         phase %in% c(1, 4, 5)) %>%
  group_by(CC.ID) %>%
  filter(sum(!is.na(mem_imm) & !is.na(mem_del)) >= 2) %>%
  ungroup()

df_mem <- df_mem %>%
  group_by(CC.ID) %>%
  mutate(A0 = as.numeric(Age[wave == '0']),
         dA = as.numeric(Age - A0))

df_mem$prac <- 1
df_mem$prac[str_detect(df_mem$wave, "0")] <- 0
df_mem$prac <- factor(df_mem$prac)
df_mem$format <- factor(df_mem$format)
```

```

df_mem$format <- ifelse(df_mem$wave == 0 & is.na(df_mem$format), 1, df_mem$format)

CC.ID <- df_mem$CC.ID[df_mem$wave == 0]
C0 <- df_mem$mem_imm[df_mem$wave == 0]
C1 <- df_mem$mem_imm[df_mem$wave == 1]
C2 <- df_mem$mem_imm[df_mem$wave == 2]
C0_1 <- df_mem$mem_del[df_mem$wave == 0]
C1_1 <- df_mem$mem_del[df_mem$wave == 1]
C2_1 <- df_mem$mem_del[df_mem$wave == 2]
educ <- df_mem$education_years[df_mem$wave == 0]
sex <- df_mem$sex[df_mem$wave == 0]
A0 <- df_mem$Age[df_mem$wave == 0]

feature_matrix <- data.frame(CC.ID, C0, C1, C2, C0_1, C1_1, C2_1, A0, sex, educ)

set.seed(987)
iso <- isolation.forest(feature_matrix[, c("C0", "C1", "C2", "C0_1", "C1_1", "C2_1", "A0")], ntrees = 1000)

feature_matrix$pred <- predict(iso, feature_matrix[, c("C0", "C1", "C2", "C0_1", "C1_1", "C2_1", "A0")])
outliers <- feature_matrix[feature_matrix$pred > 0.60, c("CC.ID", "C0", "C1", "C2", "C0_1", "C1_1", "C2_1", "A0", "sex", "educ")]
outliers_mem <- outliers

plot_outliers(df_mem, "Age", "mem_imm", "Age", "Memory Imm", min(df_mem$mem_imm), max(df_mem$mem_imm), c("C0", "C1", "C2", "C0_1", "C1_1", "C2_1", "A0", "sex", "educ"))

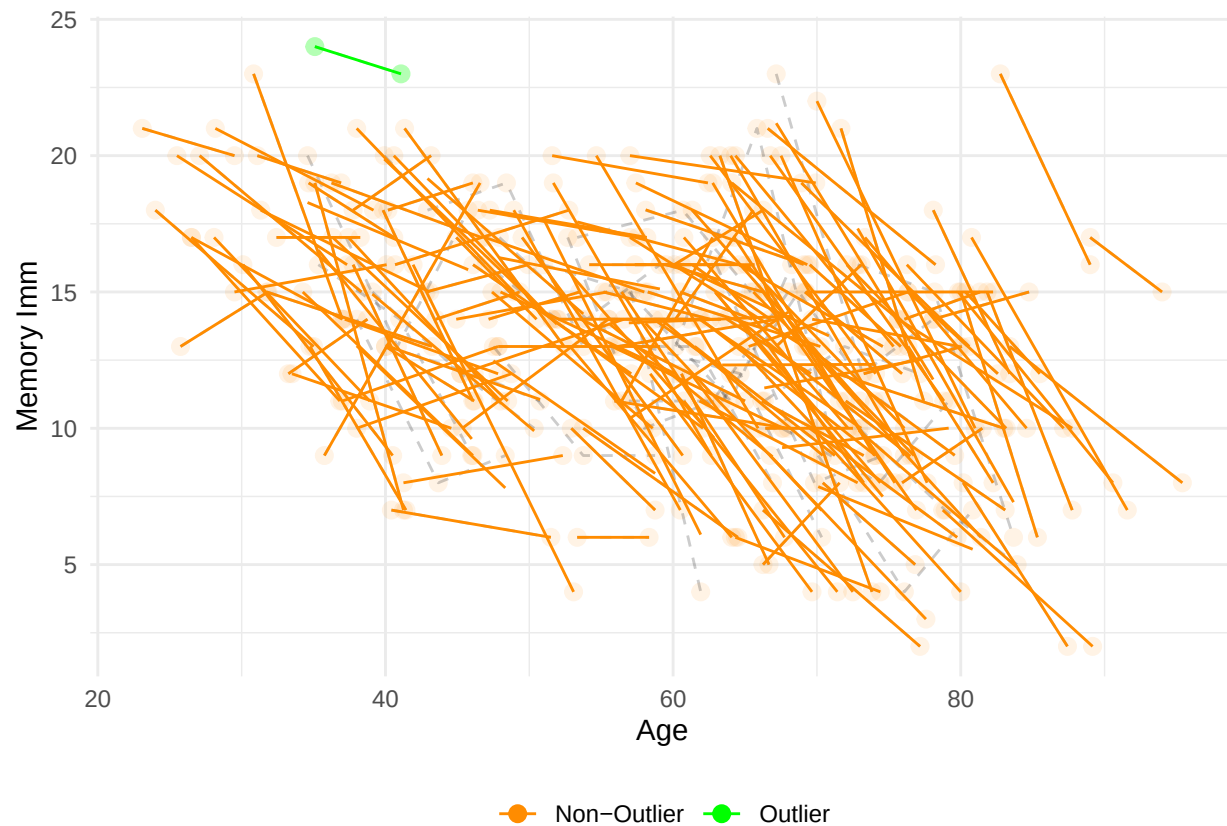
## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 157 rows containing non-finite outside the scale range
## ('stat_smooth()').

## Warning: Removed 157 rows containing missing values or values outside the scale range
## ('geom_point()').

## Warning: Removed 88 rows containing missing values or values outside the scale range
## ('geom_line()').

```



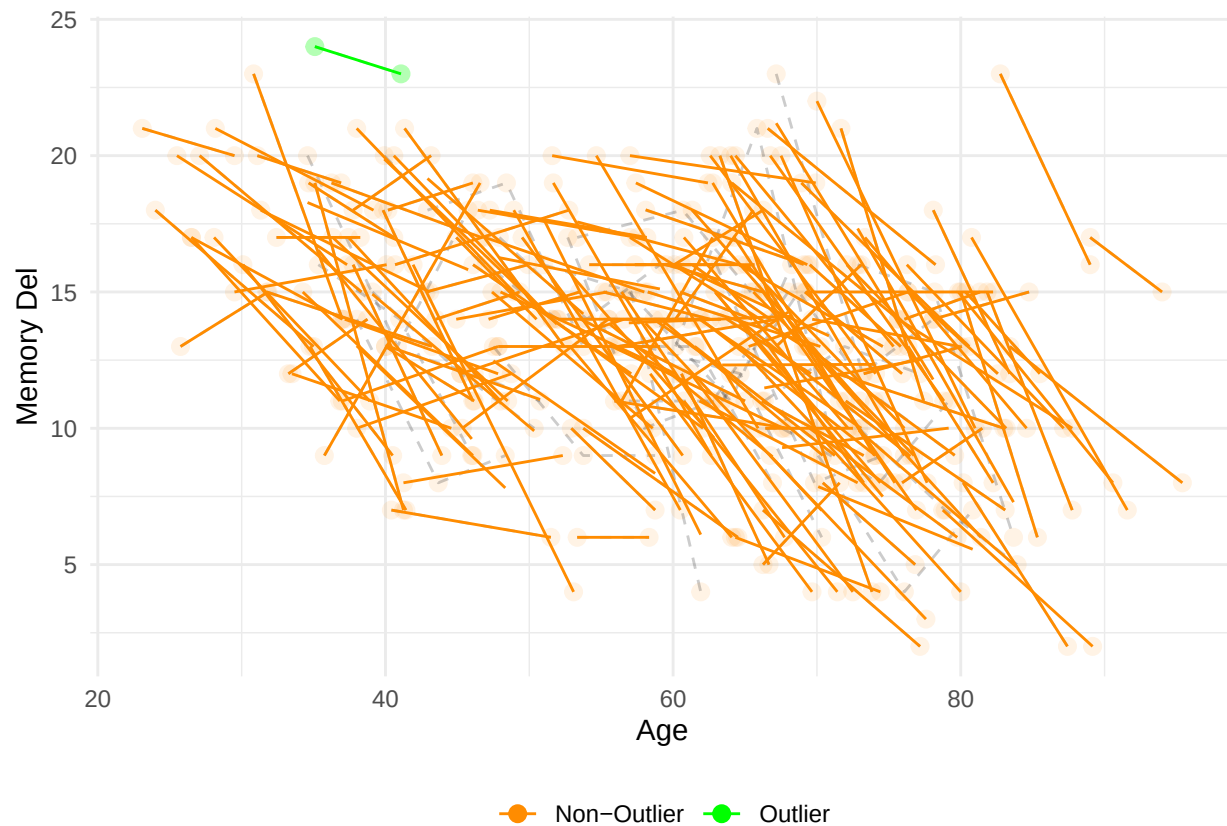
```
plot_outliers(df_mem, "Age", "mem_imm", "Age", "Memory Del", min(df_mem$mem_del), max(df_mem$mem_del),
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 157 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 88 rows containing missing values or values outside the scale range
## ('geom_line()').
```



```
df_mem$mem_imm[df_mem$CC.ID %in% outliers_mem$CC.ID] <- NA
df_mem$mem_del[df_mem$CC.ID %in% outliers_mem$CC.ID] <- NA
```

```
m_imm = lmer('scale(mem_imm) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + format + prac +
summary(m_imm)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## "scale(mem_imm) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + format + prac +
## Data: df_mem
##
## REML criterion at convergence: 982.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.47473 -0.57057  0.04635  0.53628  2.45701
##
## Random effects:
## Groups Name Variance Std.Dev.
## CC.ID (Intercept) 0.2988 0.5467
## Residual 0.4095 0.6399
## Number of obs: 399, groups: CC.ID, 185
##
## Fixed effects:
```

```

##                                Estimate Std. Error      df t value
## (Intercept)                   -0.14056    0.30975 337.32078  -0.454
## scale(A0)                     -0.25989    0.29667 338.24662  -0.876
## scale(dA)                     -0.23610    0.22842 261.34269  -1.034
## groupG3                       0.02204    0.15765 346.43592   0.140
## scale(education_years)         0.23952    0.08138 176.56351   2.943
## sex2                          0.19388    0.10946 173.08829   1.771
## format                       0.13283    0.19693 314.75061   0.675
## prac1                        -0.18478    0.28486 274.34054  -0.649
## scale(A0):scale(dA)           -0.01123    0.25256 276.02603  -0.044
## scale(A0):groupG3             0.05186    0.16271 365.89241   0.319
## scale(dA):groupG3            -0.11030    0.13748 235.63081  -0.802
## scale(A0):scale(education_years) -0.03930    0.08962 179.07796  -0.439
## scale(A0):sex2                0.04040    0.11094 179.13445   0.364
## scale(education_years):sex2   -0.05543    0.12279 177.56663  -0.451
## scale(A0):prac1              -0.04377    0.31534 282.91966  -0.139
## scale(A0):scale(dA):groupG3   -0.07352    0.14004 248.30144  -0.525
## scale(A0):scale(education_years):sex2 0.07438    0.12215 179.77794   0.609
##                                Pr(>|t|)
## (Intercept)                   0.65027
## scale(A0)                     0.38164
## scale(dA)                     0.30229
## groupG3                       0.88891
## scale(education_years)         0.00368 **
## sex2                          0.07828 .
## format                       0.50048
## prac1                        0.51709
## scale(A0):scale(dA)           0.96457
## scale(A0):groupG3             0.75009
## scale(dA):groupG3            0.42318
## scale(A0):scale(education_years) 0.66151
## scale(A0):sex2                0.71616
## scale(education_years):sex2   0.65224
## scale(A0):prac1              0.88971
## scale(A0):scale(dA):groupG3   0.60006
## scale(A0):scale(education_years):sex2 0.54331
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it

```

```

wide <- pivot_wider(df_mem[df_mem$group == "G3" & df_mem$wave %in% c(0, 2)],
  id_cols = CC.ID,
  names_from = wave,
  values_from = mem_imm)
mat <- wide[, -1]
icc(mat, model = "twoway", type = "consistency", unit = "single")

```

```

## Single Score Intraclass Correlation
##

```

```
## Model: twoway
## Type : consistency
##
## Subjects = 130
## Raters = 2
## ICC(C,1) = 0.472
##
## F-Test, H0: r0 = 0 ; H1: r0 > 0
## F(129,129) = 2.79 , p = 6.09e-09
##
## 95%-Confidence Interval for ICC Population Values:
## 0.327 < ICC < 0.596
```

```
wide <- pivot_wider(df_mem[df_mem$group == "G3" & df_mem$wave %in% c(0, 2)],
  id_cols = CC.ID,
  names_from = wave,
  values_from = mem_del)
mat <- wide[, -1]
icc(mat, model = "twoway", type = "consistency", unit = "single")
```

```
## Single Score Intraclass Correlation
##
## Model: twoway
## Type : consistency
##
## Subjects = 130
## Raters = 2
## ICC(C,1) = 0.564
##
## F-Test, H0: r0 = 0 ; H1: r0 > 0
## F(129,129) = 3.59 , p = 1.12e-12
##
## 95%-Confidence Interval for ICC Population Values:
## 0.435 < ICC < 0.671
```

```
# #sensitivity check
# df_sub <- df_mem[df_mem$group == "G3", ]
# mod <- lmer(mem_imm ~ 1 + (1 | CC.ID), data = df_sub)
# summary(mod)
# vc <- as.data.frame(VarCorr(mod))
# ICC_C <- vc$vcov[vc$grp == "CC.ID"] /
# (vc$vcov[vc$grp == "CC.ID"] + vc$vcov[vc$grp == "Residual"])
# ICC_C
#
# #sensitivity check
# mod <- lmer(mem_imm ~ 1 + (1 | CC.ID), data = df_mem)
# summary(mod)
# vc <- as.data.frame(VarCorr(mod))
# var_between <- vc$vcov[vc$grp == "CC.ID"]
# var_within <- vc$vcov[vc$grp == "Residual"]
# ICC_C <- var_between / (var_between + var_within)
# ICC_C
```


SPEED (inverse Reaction Time)

```
# outliers RT
outliers_rt_simple_R <- boxplot.stats(mydata1_R$simple_speed)$out
outliers_rt_simple_NR <- boxplot.stats(mydata1_NR$simple_speed)$out

mydata1_R_NR$simple_speed[mydata1_R_NR$return_code == 1 & mydata1_R_NR$simple_speed %in% outliers_rt_simple_R] <- NA
mydata1_R_NR$simple_speed[mydata1_R_NR$return_code == 0 & mydata1_R_NR$simple_speed %in% outliers_rt_simple_R] <- NA

outliers_rt_choice_R <- boxplot.stats(mydata1_R$choice_speed)$out
outliers_rt_choice_NR <- boxplot.stats(mydata1_NR$choice_speed)$out

mydata1_R_NR$choice_speed[mydata1_R_NR$return_code == 1 & mydata1_R_NR$choice_speed %in% outliers_rt_choice_R] <- NA
mydata1_R_NR$choice_speed[mydata1_R_NR$return_code == 0 & mydata1_R_NR$choice_speed %in% outliers_rt_choice_R] <- NA

wilcox.test(simple_speed ~ return_code, data = mydata1_R_NR)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: simple_speed by return_code
## W = 34850, p-value = 0.01523
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(choice_speed ~ return_code, data = mydata1_R_NR)
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: choice_speed by return_code
## W = 35341, p-value = 0.02333
## alternative hypothesis: true location shift is not equal to 0
```

Speed Selectivity

```
# selectivity age-driven
(mean(mydata1_R$simple_speed, na.rm = TRUE) - (mean(mydata1_R_NR$simple_speed, na.rm = TRUE)))/sd(mydata1_R$simple_speed)

## [1] 0.1648978

(mean(mydata1_R$choice_speed, na.rm = TRUE) - (mean(mydata1_R_NR$choice_speed, na.rm = TRUE)))/sd(mydata1_R$choice_speed)

## [1] 0.1679166

# selectivity age-orthogonal
clean_data <- na.omit(mydata1_R_NR[, c("simple_speed", "education_years", "return_code", "Age", "sex")])

rt_model <- lm(simple_speed ~ poly(Age,2), data = clean_data)
```

```

clean_data$regressed_simple <- residuals(rt_model, na.rm = TRUE)
returners_residuals <- clean_data$regressed_simple[clean_data$return_code == "1"]
cc700_residuals <- clean_data$regressed_simple
(mean(returners_residuals) - mean(cc700_residuals)) / sd(cc700_residuals)

```

```
## [1] 0.1784335
```

```

rt_m <-lm(simple_speed ~ return_code*poly(Age,2)*scale(education_years) , data = clean_data)
summary(rt_m)

```

```

##
## Call:
## lm(formula = simple_speed ~ return_code * poly(Age, 2) * scale(education_years),
##     data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.23172 -0.27279  0.04253  0.30634  1.16184
##
## Coefficients:
##                                     Estimate Std. Error t value
## (Intercept)                        2.80118     0.01821 153.786
## return_code1                        0.01800     0.07865   0.229
## poly(Age, 2)1                      -4.01975     0.46436  -8.657
## poly(Age, 2)2                      -0.68123     0.47288  -1.441
## scale(education_years)              0.08284     0.01857   4.462
## return_code1:poly(Age, 2)1          1.58479     2.30148   0.689
## return_code1:poly(Age, 2)2          0.31077     2.42710   0.128
## return_code1:scale(education_years) 0.05620     0.10764   0.522
## poly(Age, 2)1:scale(education_years) -0.88565     0.48564  -1.824
## poly(Age, 2)2:scale(education_years) -0.55513     0.45980  -1.207
## return_code1:poly(Age, 2)1:scale(education_years) -1.41764     3.23654  -0.438
## return_code1:poly(Age, 2)2:scale(education_years) -0.36913     3.36720  -0.110
##                                     Pr(>|t|)
## (Intercept)                        < 2e-16 ***
## return_code1                        0.8190
## poly(Age, 2)1                      < 2e-16 ***
## poly(Age, 2)2                      0.1502
## scale(education_years)             9.49e-06 ***
## return_code1:poly(Age, 2)1          0.4913
## return_code1:poly(Age, 2)2          0.8982
## return_code1:scale(education_years) 0.6017
## poly(Age, 2)1:scale(education_years) 0.0686 .
## poly(Age, 2)2:scale(education_years) 0.2277
## return_code1:poly(Age, 2)1:scale(education_years) 0.6615
## return_code1:poly(Age, 2)2:scale(education_years) 0.9127
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.422 on 687 degrees of freedom
## Multiple R-squared:  0.1725, Adjusted R-squared:  0.1592
## F-statistic: 13.02 on 11 and 687 DF, p-value: < 2.2e-16

```

```
car::Anova(rt_m, type=3, statistic = F)
```

```
## Anova Table (Type III tests)
##
## Response: simple_speed
##
##              Sum Sq Df    F value    Pr(>F)
## (Intercept)    4211.0  1 23650.2553 < 2.2e-16
## return_code      0.0  1    0.0524   0.81900
## poly(Age, 2)     13.7  2   38.3732 < 2.2e-16
## scale(education_years)  3.5  1   19.9086  9.49e-06
## return_code:poly(Age, 2)  0.1  2    0.2513   0.77786
## return_code:scale(education_years)  0.0  1    0.2726   0.60174
## poly(Age, 2):scale(education_years)  1.0  2    2.8872   0.05641
## return_code:poly(Age, 2):scale(education_years)  0.0  2    0.0974   0.90719
## Residuals      122.3 687
##
## (Intercept)          ***
## return_code
## poly(Age, 2)          ***
## scale(education_years) ***
## return_code:poly(Age, 2)
## return_code:scale(education_years)
## poly(Age, 2):scale(education_years) .
## return_code:poly(Age, 2):scale(education_years)
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# selectivity age-orthogonal choice
```

```
clean_data <- na.omit(mydata1_R_NR[, c("choice_speed", "education_years", "return_code", "Age")])
```

```
rt_model <- lm(choice_speed ~ poly(Age,2), data = clean_data)
clean_data$regressed_choice <- residuals(rt_model, na.rm = TRUE)
returners_residuals <- clean_data$regressed_choice[clean_data$return_code == "1"]
cc700_residuals <- clean_data$regressed_choice
(mean(returners_residuals) - mean(cc700_residuals)) / sd(cc700_residuals)
```

```
## [1] 0.197251
```

```
rt_m <-lm(choice_speed ~ return_code*poly(Age,2)*scale(education_years) , data = clean_data)
summary(rt_m)
```

```
##
## Call:
## lm(formula = choice_speed ~ return_code * poly(Age, 2) * scale(education_years),
##     data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.94910 -0.18524  0.01429  0.19418  0.78503
##
## Coefficients:
```

```
## Estimate Std. Error t value
## (Intercept) 1.80960 0.01191 151.977
## return_code1 0.07565 0.05105 1.482
## poly(Age, 2)1 -5.92328 0.30331 -19.529
## poly(Age, 2)2 0.42911 0.30914 1.388
## scale(education_years) 0.06736 0.01211 5.562
## return_code1:poly(Age, 2)1 -1.96860 1.49116 -1.320
## return_code1:poly(Age, 2)2 1.40502 1.56849 0.896
## return_code1:scale(education_years) -0.08302 0.06923 -1.199
## poly(Age, 2)1:scale(education_years) -0.85998 0.31638 -2.718
## poly(Age, 2)2:scale(education_years) -0.05743 0.30157 -0.190
## return_code1:poly(Age, 2)1:scale(education_years) 1.19969 2.06502 0.581
## return_code1:poly(Age, 2)2:scale(education_years) -2.75877 2.17007 -1.271
## Pr(>|t|)
## (Intercept) < 2e-16 ***
## return_code1 0.13884
## poly(Age, 2)1 < 2e-16 ***
## poly(Age, 2)2 0.16556
## scale(education_years) 3.82e-08 ***
## return_code1:poly(Age, 2)1 0.18722
## return_code1:poly(Age, 2)2 0.37068
## return_code1:scale(education_years) 0.23084
## poly(Age, 2)1:scale(education_years) 0.00673 **
## poly(Age, 2)2:scale(education_years) 0.84903
## return_code1:poly(Age, 2)1:scale(education_years) 0.56146
## return_code1:poly(Age, 2)2:scale(education_years) 0.20406
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2753 on 687 degrees of freedom
## Multiple R-squared: 0.4571, Adjusted R-squared: 0.4484
## F-statistic: 52.59 on 11 and 687 DF, p-value: < 2.2e-16
```

```
car::Anova(rt_m, type=3, statistic = F)
```

```
## Anova Table (Type III tests)
##
## Response: choice_speed
## Sum Sq Df F value
## (Intercept) 1750.29 1 23096.9500
## return_code 0.17 1 2.1959
## poly(Age, 2) 29.09 2 191.9520
## scale(education_years) 2.34 1 30.9351
## return_code:poly(Age, 2) 0.18 2 1.2103
## return_code:scale(education_years) 0.11 1 1.4382
## poly(Age, 2):scale(education_years) 0.59 2 3.9123
## return_code:poly(Age, 2):scale(education_years) 0.16 2 1.0744
## Residuals 52.06 687
## Pr(>F)
## (Intercept) < 2.2e-16 ***
## return_code 0.13884
## poly(Age, 2) < 2.2e-16 ***
## scale(education_years) 3.824e-08 ***
## return_code:poly(Age, 2) 0.29875
```

```
## return_code:scale(education_years)          0.23084
## poly(Age, 2):scale(education_years)         0.02044 *
## return_code:poly(Age, 2):scale(education_years) 0.34206
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Speed ICC

```
df_speed <- mydata %>%
  filter(group %in% c("G2", "G3"),
         phase %in% c(1, 4, 5)) %>%
  group_by(CC.ID) %>%
  filter(sum(!is.na(simple_speed) & !is.na(choice_speed)) >= 2) %>%
  ungroup()

df_speed <- df_speed %>%
  group_by(CC.ID) %>%
  mutate(A0 = as.numeric(Age[wave == '0']),
         dA = as.numeric(Age - A0))

df_speed$prac <- 1
df_speed$prac[str_detect(df_speed$wave, "0")] <- 0
df_speed$prac <- factor(df_speed$prac)
df_speed$format <- factor(df_speed$format)
df_speed$format <- ifelse(df_speed$wave == 0 & is.na(df_speed$format), 1, df_speed$format)

CC.ID <- df_speed$CC.ID[df_speed$wave == 0]
C0 <- df_speed$simple_speed[df_speed$wave == 0]
C1 <- df_speed$simple_speed[df_speed$wave == 1]
C2 <- df_speed$simple_speed[df_speed$wave == 2]
C0_1 <- df_speed$choice_speed[df_speed$wave == 0]
C1_1 <- df_speed$choice_speed[df_speed$wave == 1]
C2_1 <- df_speed$choice_speed[df_speed$wave == 2]
educ <- df_speed$education_years[df_speed$wave == 0]
sex <- df_speed$sex[df_speed$wave == 0]
A0 <- df_speed$Age[df_speed$wave == 0]

feature_matrix <- data.frame(CC.ID, C0, C1, C2, C0_1, C1_1, C2_1, A0, sex, educ)

set.seed(987)
iso <- isolation.forest(feature_matrix[, c("C0", "C1", "C2", "C0_1", "C1_1", "C2_1", "A0")], ntrees = 1000)

feature_matrix$pred <- predict(iso, feature_matrix[, c("C0", "C1", "C2", "C0_1", "C1_1", "C2_1", "A0")])
outliers <- feature_matrix[feature_matrix$pred > 0.60, c("CC.ID", "C0", "C1", "C2", "C0_1", "C1_1", "C2_1", "A0", "sex", "educ")]
outliers_speed <- outliers

plot_outliers(df_speed, "Age", "simple_speed", "Age", "Simple Speed", min(df_speed$simple_speed), max(df_speed$simple_speed))

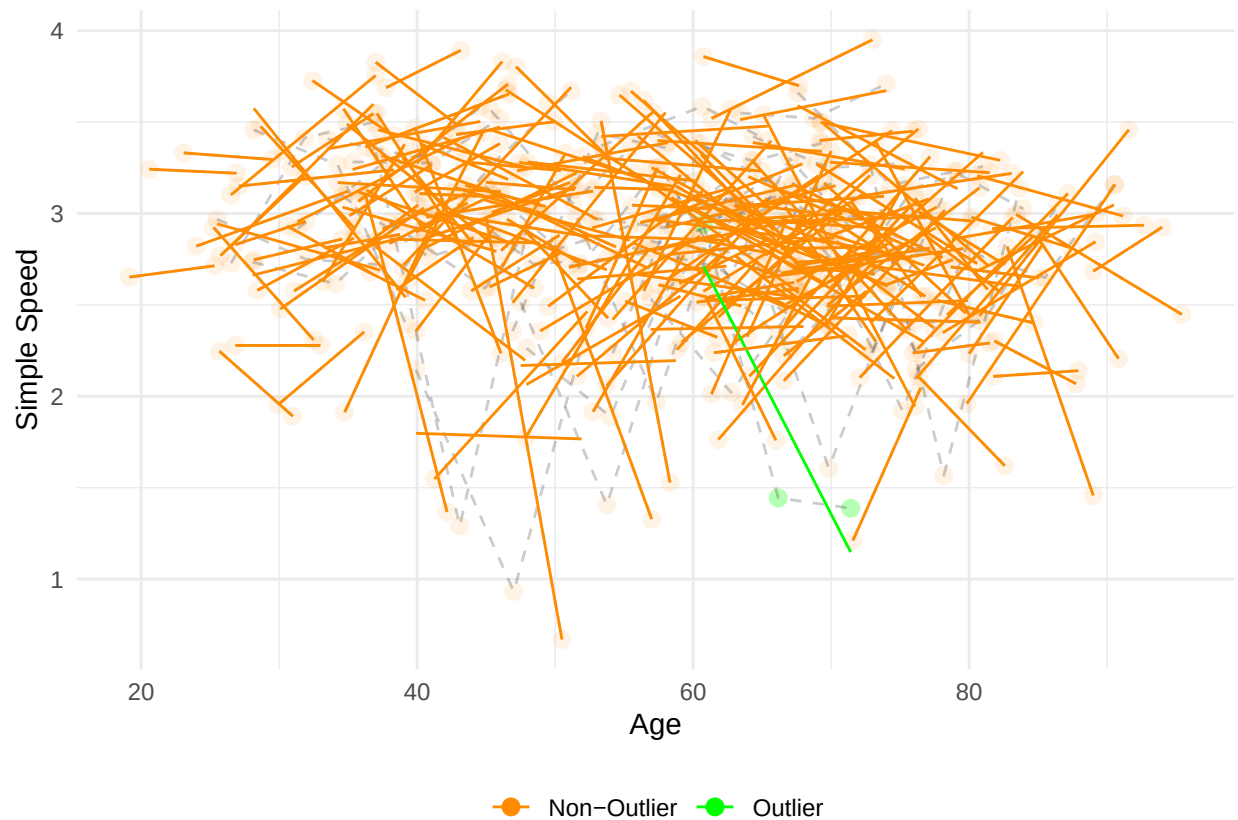
## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 207 rows containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```

```
## Warning: Removed 207 rows containing missing values or values outside the scale range  
## ('geom_point()').
```

```
## Warning: Removed 163 rows containing missing values or values outside the scale range  
## ('geom_line()').
```



```
plot_outliers(df_speed, "Age", "choice_speed", "Age", "Choice Speed", min(df_speed$choice_speed), max(d
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 210 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

```
## Warning: Removed 210 rows containing missing values or values outside the scale range  
## ('geom_point()').
```

```
## Warning: Removed 164 rows containing missing values or values outside the scale range  
## ('geom_line()').
```



```
df_speed$simple_speed[df_speed$CC.ID %in% outliers_speed$CC.ID] <- NA
df_speed$choice_speed[df_speed$CC.ID %in% outliers_speed$CC.ID] <- NA

m_simple = lmer('scale(simple_speed) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex
summary(m_simple)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## "scale(simple_speed) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + format + p
## Data: df_speed
##
## REML criterion at convergence: 1546.1
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -4.3770 -0.4813 0.1005 0.5676 1.8831
##
## Random effects:
## Groups Name Variance Std.Dev.
## CC.ID (Intercept) 0.2240 0.4733
## Residual 0.7129 0.8443
## Number of obs: 552, groups: CC.ID, 253
##
## Fixed effects:
```

```

##               Estimate Std. Error      df t value
## (Intercept)    5.287e-01  2.787e-01  4.945e+02   1.897
## scale(A0)      -2.823e-01  2.411e-01  4.887e+02  -1.171
## scale(dA)       5.247e-01  2.057e-01  4.047e+02   2.551
## groupG3        -7.986e-02  1.243e-01  4.471e+02  -0.642
## scale(education_years)  1.129e-01  7.858e-02  2.561e+02   1.436
## sex2           -2.284e-01  9.830e-02  2.509e+02  -2.324
## format          1.523e-01  1.871e-01  5.048e+02   0.814
## prac1          -9.277e-01  2.696e-01  4.212e+02  -3.441
## scale(A0):scale(dA)  -4.830e-02  2.196e-01  4.105e+02  -0.220
## scale(A0):groupG3    8.520e-02  1.289e-01  4.498e+02   0.661
## scale(dA):groupG3   -7.908e-02  1.235e-01  3.625e+02  -0.640
## scale(A0):scale(education_years) -6.539e-02  7.999e-02  2.587e+02  -0.817
## scale(A0):sex2       6.072e-02  9.955e-02  2.524e+02   0.610
## scale(education_years):sex2  3.482e-02  1.067e-01  2.594e+02   0.326
## scale(A0):prac1     -5.148e-04  2.878e-01  4.187e+02  -0.002
## scale(A0):scale(dA):groupG3  7.827e-02  1.255e-01  3.730e+02   0.624
## scale(A0):scale(education_years):sex2  9.739e-02  1.077e-01  2.619e+02   0.904
##               Pr(>|t|)
## (Intercept)    0.058411 .
## scale(A0)       0.242296
## scale(dA)       0.011100 *
## groupG3         0.520993
## scale(education_years)  0.152123
## sex2            0.020943 *
## format          0.416072
## prac1           0.000636 ***
## scale(A0):scale(dA)  0.825991
## scale(A0):groupG3    0.508899
## scale(dA):groupG3    0.522271
## scale(A0):scale(education_years)  0.414404
## scale(A0):sex2       0.542490
## scale(education_years):sex2  0.744482
## scale(A0):prac1      0.998574
## scale(A0):scale(dA):groupG3  0.533336
## scale(A0):scale(education_years):sex2  0.366657
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)      if you need it

```

```

m_choice = lmer('scale(choice_speed) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex
summary(m_choice)

```

```

## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## "scale(choice_speed) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + format + p
##      Data: df_speed
##

```



```

## REML criterion at convergence: 1270.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.8069 -0.4934  0.0546  0.5623  1.9303
##
## Random effects:
##   Groups   Name      Variance Std.Dev.
##   CC.ID    (Intercept) 0.2798   0.5290
##   Residual                0.3447   0.5871
## Number of obs: 549, groups:  CC.ID, 253
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)      0.20336    0.20881 471.07258   0.974
## scale(A0)        -0.70391    0.18317 478.63889  -3.843
## scale(dA)         0.09072    0.15009 359.36435   0.604
## groupG3           0.03592    0.10349 411.59523   0.347
## scale(education_years) 0.11852    0.06984 247.38665   1.697
## sex2              -0.08867    0.08746 244.25091  -1.014
## format            0.18613    0.13962 442.02847   1.333
## prac1            -0.67185    0.19847 370.47235  -3.385
## scale(A0):scale(dA) -0.07713    0.16141 362.77159  -0.478
## scale(A0):groupG3   0.01455    0.10744 420.07799   0.135
## scale(dA):groupG3   0.04453    0.08835 332.94876   0.504
## scale(A0):scale(education_years) -0.03590    0.07101 247.90000  -0.506
## scale(A0):sex2      0.17453    0.08855 244.88345   1.971
## scale(education_years):sex2 0.05000    0.09468 248.75927   0.528
## scale(A0):prac1     0.05741    0.21270 367.54567   0.270
## scale(A0):scale(dA):groupG3 0.07926    0.09056 340.41478   0.875
## scale(A0):scale(education_years):sex2 0.08785    0.09547 249.42848   0.920
##
##              Pr(>|t|)
## (Intercept)      0.330594
## scale(A0)         0.000138 ***
## scale(dA)         0.545918
## groupG3           0.728688
## scale(education_years) 0.090943 .
## sex2              0.311678
## format            0.183175
## prac1            0.000787 ***
## scale(A0):scale(dA) 0.633031
## scale(A0):groupG3  0.892331
## scale(dA):groupG3  0.614582
## scale(A0):scale(education_years) 0.613591
## scale(A0):sex2     0.049849 *
## scale(education_years):sex2 0.597870
## scale(A0):prac1    0.787386
## scale(A0):scale(dA):groupG3 0.382082
## scale(A0):scale(education_years):sex2 0.358347
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 17 > 12.

```

```
## Use print(x, correlation=TRUE) or
##     vcov(x)           if you need it
```

```
wide <- pivot_wider(df_speed[df_speed$group == "G3" & df_speed$wave %in% c(0, 2)],
  id_cols = CC.ID,
  names_from = wave,
  values_from = simple_speed)
mat <- wide[, -1]
icc(mat, model = "twoway", type = "consistency", unit = "single")
```

```
## Single Score Intraclass Correlation
##
## Model: twoway
## Type : consistency
##
## Subjects = 133
## Raters = 2
## ICC(C,1) = 0.155
##
## F-Test, H0: r0 = 0 ; H1: r0 > 0
## F(132,132) = 1.37 , p = 0.0373
##
## 95%-Confidence Interval for ICC Population Values:
## -0.015 < ICC < 0.316
```

```
wide <- pivot_wider(df_speed[df_speed$group == "G3" & df_speed$wave %in% c(0, 2)],
  id_cols = CC.ID,
  names_from = wave,
  values_from = choice_speed)
mat <- wide[, -1]
icc(mat, model = "twoway", type = "consistency", unit = "single")
```

```
## Single Score Intraclass Correlation
##
## Model: twoway
## Type : consistency
##
## Subjects = 132
## Raters = 2
## ICC(C,1) = 0.663
##
## F-Test, H0: r0 = 0 ; H1: r0 > 0
## F(131,131) = 4.93 , p = 1.83e-18
##
## 95%-Confidence Interval for ICC Population Values:
## 0.555 < ICC < 0.749
```